CASE STUDY

Other State-Linked Threat Actors: The Public Record

A TeamStarWolf case study. Published 3 November 2026; every source checked as of that date.

Executive summary

Most of the people on the receiving end of state-linked hacking never read about China, Russia, Iran or North Korea in the reports that describe them. They are Emirati journalists and dissidents who opened a document that asked them to enable macros. They are students at Indian universities who received macro-laden documents. They are foreign diplomats in Minsk whose Windows Update prompt was not from Microsoft. They are foreign ministries in the Middle East and North Africa whose DNS records were quietly pointed somewhere else. They are Vietnamese activists abroad, Pakistani defense researchers, users tied to the Kurdish military in Iraq and, in one Lebanese operation, thousands of people in more than 21 countries, among them medical professionals, lawyers and journalists. This study gathers the public record on the state-linked groups outside the five largest programs, dates every event, and grades every claim by the source that supports it.

The record supports four plain statements. First, this tier is real and long-running: the groups in scope have been active since at least 2004 (Darkhotel), with most first seen between 2012 and 2017, and new public reporting continued through 2025 (**Confirmed**, ATT&CK and vendor primaries [12] [26] [42] [46] [53] [54]). Second, attribution here is softer than for the big five. Only one group in the set, Dark Caracal, is attributed by ATT&CK to a named government body (Lebanon's General Directorate of General Security), and Lookout and EFF said the operation was believed to be run from one of its buildings in Beirut (**Confirmed** [12] [40]). Most of the others are "suspected", "linked", or carry no named state at all. Third, the tradecraft is mostly document-and-lure work: across the 20 groups this study binds, the most common techniques are a malicious file opened by a user (13 groups), a spearphishing attachment (11), tool download (10) and exploitation of an old client application such as Office (9) (**Confirmed** [32]). Fourth, the most damaging routes in this tier use no software vulnerability at all: Sea Turtle took over DNS registrar accounts and registries to hijack name resolution, and MoustachedBouncer injected a fake Windows Update at internet service provider level (**Confirmed** [42] [48]).

For a security program, three practical points follow. Every CVE the opened record names for these groups (22 in total) is in CISA's Known Exploited Vulnerabilities catalog, nine of them in Microsoft Office or its MSCOMCTL control (KEV lists CVE-2014-4114 under Windows) (**Confirmed** [56]), and all nine Office CVEs entered KEV between 3 November 2021 and 3 March 2022, in its initial backfill (**Confirmed** [56]), so a patch rule keyed only to new KEV entries misses them (**Inferred**). DNS registrar accounts, certificate issuance and traffic from untrusted networks need to be on the exposure register even though they carry no CVE (**Inferred** from [42] [44] [48]). And for healthcare, the record shows exposure but not a mission: none of these groups' ATT&CK descriptions names healthcare, and the vendor record shows hospitals, medical research faculty and medical professionals hit as part of wider regional and political targeting (**Inferred** from [1] [35] [36] [40]).

The study also gives, in Part 11, a short comparison with the five largest programs, including the famous platforms that public reporting links to the United States. That record differs in kind: no US government body has attributed any of those platforms to itself, and the sponsor link rests on leaks, a vendor statement and a rival government's accusation (**Inferred** from [57] to [67]).

> Evidence grade: the ATT&CK record, vendor reports, government advisories and the library's data files are each **Confirmed** against a primary source opened on 3 November 2026. Attribution statements are reported at the confidence each source states. Four items rest on secondary or contested sources and are graded **Reported**: the Stealth Falcon link to CVE-2025-33053 [50], the Indian press account of a Pakistan intelligence link to fake Aarogya Setu apps [37], the Vault 7 documents' authenticity [62] and China's attribution of the Northwestern Polytechnical University intrusion [67].

Key facts at a glance

Field	Value
Scope	State-linked groups outside China, Russia, Iran, North Korea and the United States; the MITRE mappings (Part 14) bind 20 groups: the 14 whose ATT&CK text names or suspects a state, a state body, a region (BITTER, "suspected South Asian") or a Hamas affiliation, or calls them likely state-sponsored, plus 6 that ATT&CK leaves stateless but that vendor primaries or ATT&CK's own overlap notes place in that tier (Confucius, Molerats, APT-C-23, PROMETHIUM, MoustachedBouncer and BlackOasis; the placement is **Inferred** where no source names a state)
Groups reviewed	27 Enterprise ATT&CK groups checked against their live pages, plus the Mobile-only `G0097` Bouncing Golf; 20 of the 27 are bound in the MITRE mappings (Part 14)
Firmest attribution	Dark Caracal (`G0070`): ATT&CK attributes it to Lebanon's General Directorate of General Security; Lookout and EFF said it was believed run from a GDGS building in Beirut [12] [40]
Nexus by region	South Asia (India-linked: Patchwork, Sidewinder, BITTER, Confucius; Pakistan-linked: Transparent Tribe, SideCopy); Vietnam (APT32); South Korea (Darkhotel); Lebanon (Dark Caracal, Volatile Cedar); UAE, unconfirmed (Stealth Falcon); Hamas or Gaza (Molerats, WIRTE, APT-C-23); Türkiye (Sea Turtle); Belarus interests (MoustachedBouncer)
Oldest and newest	Darkhotel, active since at least 2004 [26]; Stealth Falcon's reported zero-day use against a defense organization in Türkiye, March 2025 (**Reported** [50])
ATT&CK record	154 distinct Enterprise techniques linked to the 20 bound groups; 2 campaigns (`C0011`, `C0033`) [32]
Most-linked techniques	`T1204.002` Malicious File (13 groups), `T1566.001` Spearphishing Attachment (11), `T1105` Ingress Tool Transfer (10), `T1203` Exploitation for Client Execution (9) [32]
Exploited CVEs	22 named by the opened primaries and live ATT&CK pages, all 22 in CISA KEV (catalog 2026.11.03); 9 Office or Office components, 7 Sea Turtle edge and server flaws, 6 with KEV "Known" ransomware use [56]
Government and platform action	CISA Emergency Directive 19-01 on DNS tampering, 22 January 2019, closed 8 January 2026 [44]; Meta takedowns of APT32 (2020) and of Bitter APT and APT36 (2022) [35] [39]
Healthcare	No in-scope ATT&CK description names healthcare; vendor records show hospitals (APT32), molecular medicine researchers (Patchwork) and medical professionals (Dark Caracal) among wider target sets [1] [35] [36] [40]

Key findings

- **Attribution in this tier is a ladder with few top rungs.** ATT&CK attributes Dark Caracal to a named Lebanese agency, calls APT32 "suspected Vietnam-based", Sidewinder "suspected Indian", Transparent Tribe "suspected Pakistan-based" and Darkhotel "suspected South Korean", says the UAE link for Stealth Falcon "has not been confirmed", and names no state at all for Confucius, BITTER, Molerats, APT-C-23, PROMETHIUM, NEODYMIUM, MoustachedBouncer, Machete and Metador (**Confirmed** [2] to [31]). Vendors sometimes go further: Mandiant assesses APT32 as aligned with Vietnamese state interests, Meta tied APT32 to an IT company in Vietnam, and ESET assesses with medium confidence that MoustachedBouncer is aligned with Belarus's interests (**Confirmed** [34] [35] [48]).
- **South Asia is the densest cluster.** Six groups target each other's governments and militaries: India-linked Patchwork, Sidewinder, Confucius and BITTER, and Pakistan-linked Transparent Tribe and

SideCopy (**Confirmed** [3] to [9]). Gorgon Group, whose members ATT&CK suspects are Pakistan-based, mixes crime with attacks on UK, Spanish, Russian and US government bodies (**Confirmed** [10]). SideCopy's infection chain mimics Sidewinder's, and Confucius resembles Patchwork, so lookalike tradecraft is part of the picture (**Confirmed** [5] [9]).

- **Sea Turtle is the outlier with the widest blast radius.** Talos assessed with high confidence that a state-sponsored actor hijacked DNS through registrars, registries and ccTLD operators, hitting at least 40 organizations in 13 countries from about January 2017 to the first quarter of 2019, and the operator of Greece's .gr domain disclosed a compromise on 19 April 2019 (**Confirmed** [42] [43]). Hunt & Hackett tied 2021 to 2023 campaigns in the Netherlands to the same actor, and Microsoft tied 2024 and 2025 zero-day exploitation of a messaging server to Marbled Dust, a name it says overlaps Sea Turtle (**Confirmed** [45] [46]).
- **Surveillance of dissidents is a mission, not a side effect.** Stealth Falcon (Emirati journalists, activists and dissidents since 2012), Dark Caracal (thousands of victims, trojanized Signal and WhatsApp apps), APT32 (Vietnamese activists at home and abroad) and BlackOasis (a believed customer of Gamma Group, with opposition bloggers and activists among its interests) all target civil society (**Confirmed** [11] [12] [15] [35] [40] [49]).
- **Commercial spyware blurs who the actor is.** BlackOasis is believed to be a Gamma Group customer, Dark Caracal used FinFisher alongside custom tools, and Stealth Falcon's state link is circumstantial (**Confirmed** [15] [40] [49]). When the tooling is bought, the defensive signal is the commercial implant family as much as the operator (**Inferred**).
- **Old Office bugs still work.** Patchwork, BITTER, Sidewinder and Transparent Tribe exploit Office flaws from 2010 to 2018, among them CVE-2012-0158, CVE-2017-0199 and CVE-2017-11882; Kaspersky still recorded SideWinder exploiting CVE-2017-11882 in 2024 (**Confirmed** [3] [4] [6] [7] [53]).
- **Healthcare appears as collateral and as a lure.** Meta lists hospitals among APT32's business targets; Patchwork reached molecular medicine and biological sciences faculty for the first time in its late-2021 campaign; Dark Caracal compromised devices of medical professionals; Transparent Tribe spread a trojanized imitation of Aarogya Setu, India's COVID-19 tracking app and used a mental health survey as a lure (**Confirmed** [35] [36] [37] [38] [40]). The sustained healthcare harm in the government record comes from the big five (**Inferred** from Part 11).

Top recommendations

1. Patch Office on a build-currency rule, not a "new KEV entry" rule. Nine of the 22 CVEs this tier is known to exploit are Office or Office components, most from 2010 to 2018, and all nine entered KEV between 3 November 2021 and 3 March 2022, in its initial backfill, not as fresh additions (**Confirmed** [56]; the rule is **Inferred**). Track Office build age across the fleet and close the gap between installed and supported builds. See the Vulnerability Prioritization Reference.
2. Put DNS and registrar accounts on the exposure register. Enable registry lock, require multi-factor authentication on every DNS management and registrar account, audit public DNS records, and monitor Certificate Transparency logs for certificates you did not request, as CISA's ED 19-01 and Talos advised (**Confirmed** [42] [43] [44]). None of this has a CVE, so a CVE-driven program will not raise it by itself.
3. Treat untrusted networks as hostile for travelers and in-country staff. Darkhotel selected traveling executives on hotel networks, and ESET's advice for foreign organizations in Belarus is an end-to-end

encrypted VPN tunnel to a trusted network, so in-path injection has nothing to tamper with (**Confirmed** [48] [52]).

4. Block macros and watch what Office launches. Stealth Falcon's entry point was a macro document leading to a PowerShell backdoor, and malicious files and spearphishing attachments are the two most-linked techniques in the set (**Confirmed** [32] [49]). The library maps Email Filtering and Executable Allowlisting to these techniques (Part 14).
5. Control where mobile apps come from. Dark Caracal relied on trojanized Android messaging apps from a watering-hole site, and Transparent Tribe spread a trojanized imitation of Aarogya Setu, India's COVID-19 tracking app (**Confirmed** [37] [40]). Official stores only, with sideloading blocked by mobile device management, is the control. See the Mobile Security Reference.
6. Read the sponsor rung before acting on a name. A report that says "suspected" or names no state is evidence about tradecraft, not about intent. Carry the source's confidence into internal reporting.

Scope and method

This study covers every Enterprise group in ATT&CK v20.0 whose ATT&CK description names a state nexus other than China, Russia, Iran, North Korea or the United States, or calls the group likely state-sponsored, plus the groups ATT&CK places close to them. The library's group data was filtered first, and each candidate's live ATT&CK page was then opened. On 3 November 2026 the live ATT&CK group index listed 182 group IDs, against 178 in the library's Enterprise data; the four missing are `G0097` Bouncing Golf (Mobile only), `G1000` ALLANITE (ICS), `G1027` CyberAv3ngers (ICS) and `G1029` UNC788 (Mobile), and none has a generated group page (**Confirmed** [1] [32]). Of the four, only Bouncing Golf falls in this study's scope. For all 27 in-scope Enterprise groups, the live ATT&CK description matched the library's text (citations stripped), and every Enterprise technique the library lists for them appeared on the live page (**Confirmed** [2] [32]).

Vendor and government primaries were then opened for the groups with a public record beyond ATT&CK. The study draws on five kinds of source: the MITRE ATT&CK pages for groups and campaigns; vendor and research reports (Mandiant, Kaspersky, Cisco Talos, ESET, Microsoft, Citizen Lab, Lookout and EFF, Hunt & Hackett, ThreatDown and Meta); US government directives and advisories; the CISA KEV catalog and NVD; and the TeamStarWolf library's own data files. Every source in the Sources table was opened on 3 November 2026.

Each claim carries one of four grades.

Grade	Meaning
Confirmed	The primary source that published the fact (the agency, vendor, research lab or MITRE page) was opened on 3 November 2026 and says it. For an attribution, Confirmed means the source makes that attribution, at the confidence it states; it does not upgrade the source's own confidence
Reported	Only a secondary source was available, or the claim is a leak publisher's or a rival government's assertion that no independent primary confirms
Demonstrated	A public reproduction exists. No claim in this study rests on one
Inferred	This study's own reasoning from Confirmed facts, labelled where it appears

This study grades by source type (primary opened versus secondary only), the scheme the newer case studies use; under the case-studies README's original scheme, single-vendor reports would read as Reported.

Three rules govern the names. The study says which name each source uses ("Talos's Sea Turtle", "Microsoft's Marbled Dust", "Kaspersky's SideWinder") instead of merging labels. A source's own hedge travels with its claim ("suspected", "medium confidence", "not been confirmed"). And where a placement goes beyond what ATT&CK says, it is labelled: calling Confucius India-linked rests on its Patchwork overlap, and placing PROMETHIUM under Türkiye rests on victimology only (**Inferred** [5] [21]).

Some groups sit at the edge of this scope and have their home in a companion study in this series: POLONIUM, which researchers assess coordinated with actors affiliated with Iran's MOIS, belongs with the Iran record, and Winter Vivern, which ATT&CK links to Russian and Belorussian interests, belongs with the Russia record. Each gets one paragraph here. The Equation group's United States link is summarized in Part 11 and belongs to the United States study. Events with their own case studies (Stuxnet, SolarWinds, Sandworm, APT41, Log4Shell and Change Healthcare) are linked, not repeated.

Two primaries could not be opened on 3 November 2026: Check Point Research's report on Stealth Falcon's 2025 zero-day (the claim rests on The Record's account and on NVD listing the Check Point report as a reference) and the Russian FSB statement on Operation Triangulation (Part 11). Tradecraft is described at the level the advisories use, by technique name and ID; the study reproduces no commands, payload strings or exploit details.

Part 1: the inventory, and how firmly each group is attributed

The table lists every group reviewed, with ATT&CK's own attribution wording, the year ATT&CK gives for the start of activity, the targets ATT&CK names, and the library's Enterprise technique and software counts. Every row is **Confirmed** against the live ATT&CK page in the source column and the library data [32].

Group	ATT&CK ID	Aliases (ATT&CK)	Attribution wording in ATT&CK	Active since	Targets named by ATT&CK	Techniques / software	Source
Dark Caracal	G0070	none	Attributed to the Lebanese General Directorate of General Security (GDGS)	2012	Not stated in the description	12 / 3 (plus Mobile software)	[12]
Volatile Cedar	G0123	Lebanese Cedar	Lebanese group with political and ideological motives	2012	Individuals, companies and institutions worldwide	5 / 2	[13]
POLONIUM	G1005	Plaid Rain	Lebanon-based; researchers assess coordination with MOIS-affiliated actors	February 2022	Israeli critical manufacturing, IT and defense industry	7 / 2	[14]
APT32	G0050	SeaLotus, OceanLotus, APT-C-00, Canvas Cyclone, BISMUTH	Suspected Vietnam-based	2014	Private sector, foreign governments, dissidents, journalists; Vietnam, the Philippines, Laos, Cambodia	78 / 15	[2]

Group	ATT&CK ID	Aliases (ATT&CK)	Attribution wording in ATT&CK	Active since	Targets named by ATT&CK	Techniques / software	Source
Patchwork	G0040	Hangover Group, Dropping Elephant, Chinastrats, MONSOON, Operation Hangover	Not definitively attributed; circumstantial evidence suggests a pro-Indian or Indian entity	First observed December 2015	Diplomatic and government; US think tanks, March and April 2018	41 / 8	[3]
Sidewinder	G0121	T-APT-04, Rattlesnake	Suspected Indian	2012	Government, military and business in Asia, mainly Pakistan, China, Nepal, Afghanistan	30 / 1	[4]
Confucius	G0142	none	No state named; similarities to Patchwork	2013	Military, high-profile, business and government targets in South Asia	19 / 1	[5]
BITTER	G1002	T-APT-17	Suspected South Asian	2013	Government, energy, engineering; Pakistan, China, Bangladesh, Saudi Arabia	16 / 1	[6]
Transparent Tribe	G0134	COPPER FIELDSTONE, APT36, Mythic Leopard, ProjectM	Suspected Pakistan-based	2013	Diplomatic, defense and research organizations in India and Afghanistan	14 / 5	[7]
SideCopy	G1008	none	Pakistani group; infection chain mimics Sidewinder	2019	Indian and Afghan government personnel	16 / 2	[9]
Gorgon Group	G0078	none	Members suspected Pakistan-based or with Pakistan connections; criminal and targeted attacks	Not stated	Government organizations in the UK, Spain, Russia and the US	16 / 4	[10]
Stealth Falcon	G0038	none	Possible UAE government link on circumstantial evidence, not confirmed	2012	Emirati journalists, activists and dissidents	16 / 0	[11]
BlackOasis	G0063	none	Middle Eastern; believed customer of Gamma Group; reportedly close to NEODYMIUM	Not stated	UN figures, opposition bloggers, activists, regional correspondents, think tanks	1 / 0	[15]
Molerats	G0021	Operation Molerats, Gaza Cybergang	Arabic-speaking, politically motivated; no state named	2012	Mainly the Middle East, Europe and the US	16 / 6	[16]
WIRTE	G0090	Ashen Lepus	Believed subgroup of the Hamas-affiliated Gaza Cybergang	August 2018	Diplomatic, financial, military, legal, technology; Middle East, North Africa, Europe; wipers against Israeli targets	26 / 8	[17]
APT-C-23	G1028	Mantis, Arid Viper, Desert Falcon, TAG-63, Grey Karkadann, Big Bang APT, Two-tailed Scorpion	No sponsor named	2014; mobile spyware since 2017	The Middle East, including Israeli military assets	0 Enterprise, 3 Mobile / 1	[18]

Group	ATT&CK ID	Aliases (ATT&CK)	Attribution wording in ATT&CK	Active since	Targets named by ATT&CK	Techniques / software	Source
Bouncing Golf	G0097 (no library page)	none	A cyberespionage campaign; no sponsor named	Not stated	Middle Eastern countries	Mobile only (2)	[19]
Sea Turtle	G1041	Teal Kurma, Marbled Dust, Cosmic Wolf, SILICON	Türkiye-linked	2017	Espionage and service-provider compromise in Asia, Europe, North America; ccTLD registrars and DNS providers	27 / 1	[20]
PROMETHIUM	G0056	StrongPity	No state named; heavy emphasis on Turkish targets	2012	Global, with Turkish emphasis	11 / 2	[21]
NEODYMIUM	G0055	none	No state named; resembles PROMETHIUM; reportedly close to BlackOasis	May 2016 campaign	Heavily Turkish victims	0 / 1	[23]
MoustachedBouncer	G1019	none	No sponsor named in ATT&CK text	2014	Foreign embassies in Belarus	8 / 3	[24]
Winter Vivern	G1035	TA473, UAC-0114	Linked to Russian and Belorussian interests	2020	European government and NGO entities; sporadic Indian and US victims	27 / 0	[25]
Darkhotel	G0012	DUBNIUM, Zigzag Hail	Suspected South Korean	2004	East Asian victims, including traveling executives over hotel networks	24 / 0	[26]
Higaisa	G0126	none	Suspected South Korean origins	As early as 2009; disclosed early 2019	Government, public and trade organizations in North Korea; also China, Japan, Russia, Poland	28 / 3	[27]
The White Company	G0089	none	Likely state-sponsored; state not named	2017 to 2018 (Operation Shaheen)	Government and military organizations in Pakistan	7 / 2	[28]
Machete	G0095	APT-C-43, El Machete	Suspected Spanish-speaking; no state named	2010	Government, intelligence, military, telecom, power; Latin America (especially Venezuela), the US, Europe, Russia, Asia	11 / 1	[29]
Metador	G1013	none	Suspected cyber espionage group; no state named	First reported September 2022	A small number of telecoms, ISPs and universities in the Middle East and Africa	9 / 2	[30]
Equation	G0020	none	No sponsor named	Not stated	Not stated; zero-day use and hard-disk firmware overwrite	4 / 0	[31]

Reading the ladder. One group (Dark Caracal) is attributed to a named agency. Five carry a "suspected" country (APT32, Sidewinder, Transparent Tribe, Darkhotel, Higaisa) and three a country link without the hedge (SideCopy "Pakistani", Volatile Cedar "Lebanese", Sea Turtle "Türkiye-linked"). One is explicitly unconfirmed (Stealth Falcon). The rest name a region, a language, a movement or nothing (**Inferred** from the table). That

spread is the main difference between this tier and the big five, where governments publish joint attributions with named units (Part 11).

The healthcare question starts here. None of these 28 descriptions names healthcare as a target. The two ATT&CK groups whose descriptions do mention healthcare, G0107 Whitefly (linked to the SingHealth attack) and G0071 Orangeworm (healthcare targeting, likely corporate espionage), are outside this scope because ATT&CK names no sponsor for either (**Inferred** from [1]). The vendor record in Parts 2 to 5 and Part 10 fills in what ATT&CK does not.

Part 2: South Asia

India-linked groups

Patchwork (G0040) is the best documented of the India-linked groups. ATT&CK says it is not definitively attributed but that circumstantial evidence suggests a pro-Indian or Indian entity; it was first observed in December 2015, targets diplomatic and government sectors, and spearphished US think tanks in March and April 2018 (**Confirmed** [3]). ThreatDown (Malwarebytes) describes it plainly as an Indian threat actor and documented a late November to early December 2021 campaign against Pakistan's Ministry of Defense, defense and academic institutions and research institutes (**Confirmed** [36]). That campaign used RTF lures impersonating Pakistani authorities that exploited the Microsoft Equation Editor (T1566.001, T1203) to drop a BADNEWS variant ThreatDown calls Ragnatela; the post names no CVE (**Confirmed** [36]). For the first time, the targets included faculty researching molecular medicine and biological sciences (**Confirmed** [36]). ATT&CK's group page names six Office and Windows CVEs for Patchwork: CVE-2017-8570, CVE-2012-1856, CVE-2014-4114, CVE-2017-0199, CVE-2017-11882 and CVE-2015-1641 (**Confirmed** [3] [32]).

Sidewinder (G0121) is called a suspected Indian group by ATT&CK, active since at least 2012 against government, military and business entities in Asia, mainly Pakistan, China, Nepal and Afghanistan (**Confirmed** [4]). Kaspersky's October 2024 report widens the picture: government, military, logistics, infrastructure, telecom, finance, universities and oil trading across South Asia, the Middle East and Africa, reached through template-injection documents that exploit CVE-2017-11882 (T1566.001, T1203) and a post-exploitation tool Kaspersky calls StealerBot (**Confirmed** [53]). In the second half of 2024 the group sharply increased attacks on maritime infrastructure and logistics and targeted nuclear energy entities in South Asia (**Confirmed** [54]). Neither Kaspersky report names healthcare. ATT&CK lists CVE-2017-11882 and CVE-2020-0674 for the group and one software object, Koadic (S0250) (**Confirmed** [4] [32]).

Confucius (G0142) has targeted military, high-profile, business and government targets in South Asia since at least 2013, and researchers note similarities to Patchwork; ATT&CK names no state, so calling it India-linked is **Inferred** from that overlap (**Confirmed** for the ATT&CK text [5]). Its one Enterprise software object is WarzoneRAT (S0670) [32].

BITTER (G1002) is a suspected South Asian espionage group active since at least 2013 against government, energy and engineering organizations in Pakistan, China, Bangladesh and Saudi Arabia (**Confirmed** [6]). ATT&CK lists CVE-2012-0158, CVE-2017-11882, CVE-2018-0798 and CVE-2018-0802 for initial exploitation and CVE-2021-1732 for privilege escalation, and its software object ZxxZ (S1013) (**Confirmed** [6] [32]). Meta reported action against Bitter APT in its second-quarter 2022 adversarial threat report; the landing page gives no sector detail (**Confirmed** [39]).

Pakistan-linked groups

Transparent Tribe (G0134), also APT36, is called suspected Pakistan-based by ATT&CK and has targeted diplomatic, defense and research organizations in India and Afghanistan since at least 2013 (**Confirmed** [7]). Kaspersky describes it as typically targeting Indian military and government personnel (**Confirmed** [37]). In 2020 the group trojanized an Android package imitating Aarogya Setu, India's official COVID-19 tracking app, carrying a modified AhMyth remote access tool (**Confirmed** [37]). Kaspersky cites Indian press reports for a Pakistan intelligence link to fake Aarogya Setu apps sent to Indian Army groups, and for the Indian Army's warning to personnel in late April 2020; that link is **Reported** [37]. From December 2021, the group went after students at Indian universities and colleges with CrimsonRAT delivered through macro documents, a shift away from its usual government and military targets, and ATT&CK records the activity as campaign C0011, still ongoing as of July 2022 (**Confirmed** [8] [38]). One lure file in that campaign was themed as a mental health survey: a health-themed lure, not a health-sector victim (**Confirmed** [38]). Talos assesses with high confidence that a hosting provider in Lahore ran parts of the infrastructure (**Confirmed** [38]). Meta's 2022 report links APT36 to state-linked actors in Pakistan (**Confirmed** [39]). ATT&CK names CVE-2012-0158 and CVE-2010-3333 for the group (**Confirmed** [7]).

SideCopy (G1008) is a Pakistani group that has targeted South Asian countries, including Indian and Afghan government personnel, since at least 2019, with an infection chain that mimics Sidewinder's; its software objects are Action RAT (S1028) and AuTo Stealer (S1029) (**Confirmed** [9] [32]).

Gorgon Group (G0078) has members ATT&CK suspects are Pakistan-based or have Pakistan connections, and it mixes criminal campaigns with targeted attacks on government organizations in the UK, Spain, Russia and the US (**Confirmed** [10]). Because of the criminal mix it is not bound in Part 14.

The White Company (G0089) is the reverse case: ATT&CK calls it a likely state-sponsored actor but does not name the state, and from 2017 through 2018 it ran Operation Shaheen against government and military organizations in Pakistan, with NETWIRE (S0198) and Revenge RAT (S0379) (**Confirmed** [28] [32]).

What South Asia shows

The South Asian record is a contest between neighbours, conducted through documents. Lookalike chains (SideCopy copying Sidewinder, Confucius resembling Patchwork) mean that a technique match is weak evidence of who is behind an intrusion (**Inferred** from [5] [9]). For an organization outside the region, exposure comes through staff, partners and research ties to South Asian institutions rather than through direct targeting (**Inferred** from [53] and the target lists above).

Part 3: Southeast and East Asia

APT32 (Vietnam)

APT32 (G0050) has the largest ATT&CK record in this tier: 78 techniques and 15 software objects in the library's data (**Confirmed** [2] [32]). ATT&CK calls it a suspected Vietnam-based group active since at least 2014, targeting the private sector, foreign governments, dissidents and journalists, focused on Vietnam, the Philippines, Laos and Cambodia, with heavy use of strategic web compromises (**Confirmed** [2]).

Mandiant's May 2017 report assesses APT32 as aligned with Vietnamese state interests. It dates intrusions against foreign companies investing in Vietnam to at least 2014, against dissidents and journalists to at least 2013, and records the targeting of a Western legislature in 2014 (**Confirmed** [34]). The sectors Mandiant lists

are manufacturing, consumer products, hospitality, network security, technology infrastructure, banking, media, foreign governments and the Vietnamese diaspora, in Vietnam, Germany, the Philippines, China, the US and Australia; healthcare is not named (**Confirmed** [34]). Delivery was macro-document spearphishing (`T1566.001`, `T1204.002`) carrying WINDSHIELD (`S0155`), KOMPROGO (`S0156`), SOUNDBITE (`S0157`), PHOREAL (`S0158`) and Cobalt Strike BEACON (`S0154`) (**Confirmed** [34] [32]).

In December 2020 Meta linked APT32 to CyberOne Group, an IT company in Vietnam (**Confirmed** [35]). Meta's target list is wider than Mandiant's: Vietnamese activists at home and abroad, the governments of Laos and Cambodia, NGOs, news agencies, and businesses in IT, hospitality, agriculture, commodities, hospitals, retail, automotive and mobile services (**Confirmed** [35]). The methods were fake personas and Pages, malicious Play Store apps, watering holes, DLL side-loading and malicious documents (**Confirmed** [35]).

Darkhotel and Higaisa (South Korea, suspected)

Darkhotel (`G0012`) is the oldest group in the set. ATT&CK calls it a suspected South Korean group that has targeted East Asian victims since at least 2004, including traveling executives over hotel Internet networks (**Confirmed** [26]). Kaspersky's November 2014 report describes the hotel-network delivery, Flash zero-days in spearphishing, and abuse of weak certificates from at least ten certificate authorities, later replaced by stolen legitimate certificates (`T1189`, `T1203`, `T1553.002`); Kaspersky made no language or nationality attribution, so the South Korean suspicion is ATT&CK's wording, not Kaspersky's (**Confirmed** [52]). ATT&CK names CVE-2015-8651 for the group (**Confirmed** [26]).

Higaisa (`G0126`) is suspected to have South Korean origins and targets government, public and trade organizations in North Korea, as well as China, Japan, Russia and Poland. It was first disclosed in early 2019 but assessed to have operated since as early as 2009 (**Confirmed** [27]). Its suspected origin rests on ATT&CK's wording alone and no vendor primary for it was opened, so it is not bound in Part 14; its 28 techniques sit on its group page.

Part 4: the Middle East

Lebanon

Dark Caracal (`G0070`) is the clearest case of a named government body in this tier. ATT&CK attributes it to the Lebanese General Directorate of General Security and dates it to at least 2012 (**Confirmed** [12]). The January 2018 Lookout and EFF report says the operation was believed to be administered from a GDGS building in Beirut and counted thousands of victims in more than 21 countries, with hundreds of gigabytes exfiltrated (**Confirmed** [40]). The victims included military personnel, enterprises, medical professionals, activists, journalists, lawyers and educational institutions, and targets also included governments, utilities, financial institutions, manufacturers and defense contractors (**Confirmed** [40]). The main vector was trojanized Signal and WhatsApp Android apps that relied on permissions the user granted, with no exploit, and EFF named victims in the US, Canada, Germany, Lebanon and France (**Confirmed** [41]). The actor also used a watering-hole site, phishing through WhatsApp and Facebook, and physical access (`T1189`, `T1566.003`), with a toolset of Pallas, FinFisher (`S0182`), Bandook (`S0234`) and CrossRAT (`S0235`) (**Confirmed** [40] [32]). Pallas is a Mobile-only ATT&CK object with no library page.

Volatile Cedar (`G0123`) is a Lebanese group operating since 2012 with political and ideological motives against individuals, companies and institutions worldwide, with Caterpillar WebShell (`S0572`) and Explosive (`S0569`) (**Confirmed** [13] [32]).

POLONIUM (G1005) is Lebanon-based by ATT&CK's account and has targeted Israeli critical manufacturing, IT and defense-industry organizations since at least February 2022; researchers assess it coordinated with actors affiliated with Iran's MOIS, and its tools are CreepyDrive (S1023) and CreepySnail (S1024) (**Confirmed** [14] [32]). Its fuller home is the Iran record in this series.

The Gulf: Stealth Falcon and BlackOasis

Stealth Falcon (G0038) has run targeted spyware attacks against Emirati journalists, activists and dissidents since at least 2012; ATT&CK says circumstantial evidence suggests a possible UAE government link that has not been confirmed (**Confirmed** [11]). Citizen Lab's May 2016 report documented targets including a London-based journalist, delivery by macro documents and tracked links (T1566.001, T1566.002, T1204.002), a PowerShell backdoor with RC4-encrypted command and control, and theft of browser and credential-vault data (T1059.001, T1573.001, T1555.003, T1555.004) (**Confirmed** [49]). In 2025, according to Check Point as relayed by The Record, Stealth Falcon exploited CVE-2025-33053, a WebDAV and Internet Shortcut working-directory flaw, as a zero-day against a major defense organization in Türkiye in March 2025, using tools Check Point calls Horus Agent and Horus Loader (**Reported** [50]). The Check Point primary could not be opened; NVD lists it as a reference (**Confirmed** [51]). CISA added CVE-2025-33053 to KEV on 10 June 2025, with a due date of 1 July 2025 and CWE-73, and NVD published it the same day with a CVSS 3.1 score of 8.8 from Microsoft (**Confirmed** [51] [56]).

BlackOasis (G0063) is described by ATT&CK as a Middle Eastern group believed to be a customer of Gamma Group, with interests that include UN figures, opposition bloggers, activists, regional correspondents and think tanks; it is reportedly closely associated with NEODYMIUM (**Confirmed** [15]). Its ATT&CK record is one technique.

Hamas and Gaza

Three groups carry a Gaza or Hamas nexus but no state sponsor. Molerats (G0021), the Gaza Cybergang, is an Arabic-speaking, politically motivated group operating since 2012, with victims mainly in the Middle East, Europe and the US (**Confirmed** [16]). WIRTE (G0090) is believed to be a subgroup of the Hamas-affiliated Gaza Cybergang, active since at least August 2018 against diplomatic, financial, military, legal and technology organizations in the Middle East, North Africa and Europe, and it has added wiper attacks against Israeli targets (**Confirmed** [17]). APT-C-23 (G1028), also Arid Viper and Desert Falcon, has been active since at least 2014, focused on the Middle East including Israeli military assets, and has built Android and iOS spyware since 2017; the library holds no Enterprise technique and three Mobile techniques for it (**Confirmed** [18] [32]). Bouncing Golf (G0097) is a Mobile-domain cyberespionage campaign against Middle Eastern countries with no sponsor named and no library page (**Confirmed** [19]).

Part 5: Türkiye and Belarus

Sea Turtle

Sea Turtle (G1041) is called a Türkiye-linked actor by ATT&CK, active since at least 2017 in espionage and service-provider compromise across Asia, Europe and North America, and notable for targeting ccTLD registrars and DNS providers to hijack resolution and spoof login portals (**Confirmed** [20]). Its record is the most consequential in this tier.

Talos's April 2019 report assessed with high confidence that an advanced state-sponsored actor was behind the activity, without naming a nation (**Confirmed** [42]). The campaign ran from about January 2017 to the first quarter of 2019 and hit at least 40 organizations in 13 countries: primarily foreign ministries, militaries, intelligence bodies and energy organizations in the Middle East and North Africa, with telecoms, ISPs, IT firms, registrars and a registry as secondary victims; no healthcare organization appears (**Confirmed** [42]). The method was DNS record hijacking through registrar and registry compromise (`T1584.002`), adversary-in-the-middle servers that harvested credentials (`T1557`), certificates obtained for victim domains (`T1588.004`) and stolen credentials (`T1078`) (**Confirmed** [42]). Talos listed seven CVEs the actor exploited (`T1190`) and said the list is likely incomplete: CVE-2009-1151, CVE-2014-6271, CVE-2017-3881, CVE-2017-6736, CVE-2017-12617, CVE-2018-0296 and CVE-2018-7600, all now in KEV (**Confirmed** [42] [56]).

Talos's July 2019 follow-up reported that ICS-Forth, the operator of Greece's .gr ccTLD, disclosed a compromise on 19 April 2019 and that attacker access lasted through at least 24 April (**Confirmed** [43]). It recommended registry lock, multi-factor authentication for DNS management and VPN access, DNSSEC signing and validation, restricting IMAP, and passive DNS monitoring (**Confirmed** [43]).

Before either Talos report, on 22 January 2019, CISA issued Emergency Directive 19-01, ordering federal agencies to audit DNS records, change DNS account passwords, add multi-factor authentication to DNS accounts and monitor Certificate Transparency logs, each within 10 business days (**Confirmed** [44]). The directive names no actor, so linking it to Sea Turtle is **Inferred** from timing and subject. CISA closed the directive on 8 January 2026 (**Confirmed** [44]).

Hunt & Hackett's January 2024 report attributed 2021 to 2023 campaigns in the Netherlands against telecom, media, ISPs, IT service providers and Kurdish websites to Sea Turtle, which it assesses acts in line with Turkish interests (**Confirmed** [45]). Access came through compromised cPanel accounts and SSH (`T1078`, `T1133`). The actor used SnappyTCP (`S1163`), cleared logs and command history (`T1685.006`, `T1690`), and staged an email archive in a public web directory (`T1114.001`, `T1074.002`, `T1560.001`) (**Confirmed** [45]).

In May 2025 Microsoft reported that Marbled Dust, a Türkiye-affiliated espionage actor it says overlaps Sea Turtle and UNC1326, exploited CVE-2025-27920 in Output Messenger (`T1190`) against users tied to the Kurdish military in Iraq, with activity from April 2024 (**Confirmed** [46]). Microsoft says the actor historically targets European and Middle Eastern governments opposing Turkish interests, plus telecom and IT; no healthcare is named (**Confirmed** [46]).

PROMETHIUM and NEODYMIUM

PROMETHIUM (`G0056`), also StrongPity, is an espionage group active since at least 2012, operating globally with a heavy emphasis on Turkish targets; ATT&CK names no state, so a Türkiye link is **Inferred** from victimology only (**Confirmed** for the ATT&CK text [21]). Campaign `C0033` used StrongPity (`S0491`) against Android users, and ATT&CK calls it the group's first publicly documented mobile campaign; the library dates it from May 2016 to January 2023 (**Confirmed** [22] [32]). NEODYMIUM (`G0055`) ran a campaign in May 2016 that heavily targeted Turkish victims, resembles PROMETHIUM, and is reportedly closely associated with BlackOasis, though ATT&CK found no evidence the names are aliases; no state is named, and its one software object is Wingbird (`S0176`) (**Confirmed** [23] [32]).

Belarus

MoustachedBouncer (G1019) is described by ATT&CK as a cyberespionage group active since at least 2014 that targets foreign embassies in Belarus, with no sponsor named (**Confirmed** [24]). ESET's August 2023 report assesses with medium confidence that it is aligned with Belarus's interests (**Confirmed** [48]). Since 2020 it has likely tampered with traffic at ISP level to serve a fake Windows Update (T1659 Content Injection) that delivers Disco (S1088); it also uses NightClub (S1090) with SMTP command and control and SharpDisco (S1089), and a Disco plugin uses CVE-2021-1732 for privilege escalation (T1068) (**Confirmed** [48] [32]). ESET's advice is an end-to-end encrypted VPN tunnel to a trusted network (**Confirmed** [48]).

UNC1151, also known through the Ghostwriter information operations, is linked by Mandiant with high confidence to the Belarusian government and with moderate confidence to its military, without ruling out Russian contributions (**Confirmed** [47]). From 2016 to 2021 it targeted mainly Ukraine, Lithuania, Latvia, Poland and Germany: governments (defense ministries most often), webmail providers, businesses, intergovernmental bodies, dissidents and media, chiefly through credential phishing (T1566.002); no healthcare is named (**Confirmed** [47]). No UNC1151 or Ghostwriter object exists in the library's ATT&CK data, so it cannot be bound by ID (**Confirmed** [32]).

Winter Vivern (G1035) is linked by ATT&CK to Russian and Belorussian interests, active since at least 2020 against European government and NGO entities, with sporadic Indian and US victims, using document phishing and server-side exploitation (**Confirmed** [25]). Its home is the Russia record in this series.

Part 6: unnamed sponsors and Spanish-speaking groups

Three groups appear in espionage reporting with no state attached. Machete (G0095) is a suspected Spanish-speaking group active since at least 2010, focused on Latin America and especially Venezuela, plus the US, Europe, Russia and Asia, against government, intelligence, military, telecom and power organizations (**Confirmed** [29]). Metador (G1013) was first reported in September 2022 against a small number of telecoms, ISPs and universities in the Middle East and Africa; it was named after a malware string, researchers expected Spanish-language command-and-control responses, and its tools are Mafalda (S1060) and metaMain (S1059) (**Confirmed** [30] [32]). Among the groups reviewed here, The White Company (Part 2) is the only one ATT&CK calls likely state-sponsored without naming the state (**Confirmed** [28]).

These groups sit in the inventory because their targets and resourcing look like state work, but a defender should treat the sponsor question as open. They are not bound in Part 14, except The White Company.

Part 7: the operational record

One row per dated campaign, report or government action in the verified record for this tier. "Since" rows mark the start of activity as ATT&CK or the vendor states it.

Date	Group	Event or report	Targets	Confidence as the source states it	Grade	Source
2004 onward	Darkhotel	Active since at least 2004, including hotel-network targeting of traveling executives	East Asian victims	ATT&CK: suspected South Korean	Confirmed	[26]

Date	Group	Event or report	Targets	Confidence as the source states it	Grade	Source
2009 onward	Higaisa	Assessed to have operated since as early as 2009	Government, public and trade organizations in North Korea, China, Japan, Russia, Poland	ATT&CK: suspected South Korean origins	Confirmed	[27]
2010 onward	Machete	Active since at least 2010	Government, intelligence, military, telecom, power; Latin America	ATT&CK: suspected Spanish-speaking	Confirmed	[29]
2012 onward	Dark Caracal	Operating since at least 2012	Thousands of victims in 21+ countries	ATT&CK: attributed to GDGS; Lookout and EFF: believed run from a GDGS building	Confirmed	[12] [40]
2012 onward	Stealth Falcon	Targeted spyware attacks since at least 2012	Emirati journalists, activists, dissidents	ATT&CK: UAE link not confirmed	Confirmed	[11]
2012 onward	Sidewinder, Volatile Cedar, Molerats, PROMETHIUM	Each active since 2012 by ATT&CK's account	South Asia; worldwide; Middle East and Europe; Turkish emphasis	As in Part 1	Confirmed	[4] [13] [16] [21]
2013 onward	APT32	Targeting of dissidents and journalists since at least 2013	Vietnamese dissidents and journalists	Mandiant: aligned with Vietnamese state interests	Confirmed	[34]
2013 onward	Transparent Tribe, Confucius, BITTER	Each active since at least 2013	India and Afghanistan; South Asia; Pakistan, China, Bangladesh, Saudi Arabia	As in Part 1	Confirmed	[7] [5] [6]
2014	APT32	Intrusions against foreign companies investing in Vietnam; targeting of a Western legislature	Private sector; a Western legislature	Mandiant	Confirmed	[34]
2014 onward	MoustachedBouncer, APT-C-23	Each active since at least 2014	Foreign embassies in Belarus; the Middle East	ATT&CK names no sponsor for either	Confirmed	[24] [18]
2014-11-10	Darkhotel	Kaspersky report on hotel-network delivery, Flash zero-days and certificate abuse	Traveling executives	Kaspersky makes no nationality attribution	Confirmed	[52]
2015-12	Patchwork	First observed	Diplomatic and government	ATT&CK: pro-Indian or Indian entity suggested	Confirmed	[3]
2016 to 2021	UNC1151	Credential phishing and Ghostwriter-linked operations	Ukraine, Lithuania, Latvia, Poland, Germany	Mandiant: high confidence Belarusian government	Confirmed	[47]
2016-05	NEODYMIUM	Campaign heavily targeting Turkish victims	Turkish victims	No state named	Confirmed	[23]
2016-05 to 2023-01	PROMETHIUM	Campaign C0033, StrongPity against Android users	Android users	No state named	Confirmed	[22]

Date	Group	Event or report	Targets	Confidence as the source states it	Grade	Source
2016-05-29	Stealth Falcon	Citizen Lab report on macro documents and a PowerShell backdoor	Emirati dissidents; a London-based journalist	Circumstantial UAE link	Confirmed	[49]
2017 onward	APT-C-23	Android and iOS spyware development	The Middle East	No sponsor named	Confirmed	[18]
2017-01 to 2019-Q1	Sea Turtle	DNS hijacking through registrars and registries	40+ organizations in 13 countries, mainly MENA national security bodies	Talos: high confidence, state-sponsored, nation not named	Confirmed	[42]
2017 to 2018	The White Company	Operation Shaheen	Government and military organizations in Pakistan	ATT&CK: likely state-sponsored	Confirmed	[28]
2017-05-14	APT32	Mandiant report on APT32	Private sector, foreign governments, diaspora	Aligned with Vietnamese state interests	Confirmed	[34]
2018-01-18	Dark Caracal	Lookout and EFF report	Military, enterprises, medical professionals, activists, journalists, lawyers, educational institutions	Believed run from a GDGS building	Confirmed	[40] [41]
2018-03 to 2018-04	Patchwork	Spearphishing of US think tanks	US think tanks	ATT&CK	Confirmed	[3]
2018-08 onward	WIRTE	Active since at least August 2018	Middle East, North Africa, Europe	Believed Gaza Cybergang subgroup	Confirmed	[17]
2019 onward	SideCopy	Active since at least 2019	Indian and Afghan government personnel	ATT&CK: Pakistani	Confirmed	[9]
2019-01-22	(no actor named)	CISA Emergency Directive 19-01, DNS infrastructure tampering	US federal agencies' DNS	Names no actor	Confirmed	[44]
Early 2019	Higaisa	First public disclosure	As above	ATT&CK	Confirmed	[27]
2019-04-17	Sea Turtle	Talos report, seven CVEs named	As above	High confidence	Confirmed	[42]
2019-04-19 to 2019-04-24	Sea Turtle	ICS-Forth (.gr ccTLD) compromise disclosed; access through at least 24 April	The .gr registry	Talos	Confirmed	[43]
2019-07-09	Sea Turtle	Talos follow-up with hardening advice	Registries and their customers	Talos	Confirmed	[43]
2020	Transparent Tribe	Trojanized Aarogya Setu COVID-19 app with modified AhMyth	Android users (app, Confirmed); Indian Army groups (Reported via Kaspersky)	Kaspersky; the Pakistan intelligence link is from Indian press	Confirmed (app); Reported (link)	[37]

Date	Group	Event or report	Targets	Confidence as the source states it	Grade	Source
2020-04	Transparent Tribe	Indian Army warns personnel about fake Aarogya Setu apps	Indian Army	Indian press via Kaspersky	Reported	[37]
2020 onward	MoustachedBouncer	ISP-level injection of a fake Windows Update	Foreign embassies in Belarus	ESET: likely	Confirmed	[48]
2020 onward	Winter Vivern	Active since at least 2020	European government and NGOs	ATT&CK: Russian and Belorussian interests	Confirmed	[25]
2020-08-26	Transparent Tribe	Kaspersky "Transparent Tribe, part 2"	Indian military and government personnel	Kaspersky	Confirmed	[37]
2020-12-10	APT32	Meta takedown; APT32 linked to CyberOne Group	Activists, governments of Laos and Cambodia, NGOs, news agencies, businesses including hospitals	Meta	Confirmed	[35]
2021-11 to 2021-12	Patchwork	Campaign against Pakistan's Ministry of Defense and research institutes, including molecular medicine faculty (published 2022-01-07)	Pakistani defense, academic and research bodies	ThreatDown: Indian threat actor	Confirmed	[36]
2021-11-16	UNC1151	Mandiant links UNC1151 to the Belarusian government	As above	High confidence (government), moderate (military)	Confirmed	[47]
2021 to 2023	Sea Turtle	Campaigns in the Netherlands through cPanel and SSH access	Telecom, media, ISPs, IT providers, Kurdish websites	Hunt & Hackett: in line with Turkish interests	Confirmed	[45]
2021-12 to 2022-07	Transparent Tribe	Campaign C0011 against students (Talos, published 2022-07-13)	Indian universities and colleges	Talos: high confidence on Lahore hosting	Confirmed	[8] [38]
2022-02 onward	POLONIUM	Targeting of Israeli organizations	Israeli critical manufacturing, IT, defense industry	ATT&CK: Lebanon-based, coordinated with MOIS-affiliated actors	Confirmed	[14]
2022-08-04	Bitter APT, APT36	Meta second-quarter 2022 adversarial threat report	Not detailed on the landing page	Meta: APT36 linked to state-linked actors in Pakistan	Confirmed	[39]
2022-09	Metador	First reported	Telecoms, ISPs, universities in the Middle East and Africa	No state named	Confirmed	[30]
2023-08-10	MoustachedBouncer	ESET report	Foreign embassies in Belarus	Medium confidence, aligned with Belarus	Confirmed	[48]
2024-01-05	Sea Turtle	Hunt & Hackett report	As above	In line with Turkish interests	Confirmed	[45]

Date	Group	Event or report	Targets	Confidence as the source states it	Grade	Source
2024-04 onward	Marbled Dust (Sea Turtle overlap)	Exploitation of CVE-2025-27920 in Output Messenger (published 2025-05-12)	Users tied to the Kurdish military in Iraq	Microsoft: Türkiye-affiliated	Confirmed	[46]
2024-10-15	SideWinder	Kaspersky report; CVE-2017-11882 and StealerBot	Government, military, logistics, telecom, finance, universities across South Asia, the Middle East, Africa	Kaspersky	Confirmed	[53]
2024 second half	SideWinder	Sharp increase against maritime and logistics; nuclear energy targets (published 2025-03-10)	Maritime infrastructure; South Asian nuclear energy	Kaspersky	Confirmed	[54]
2025-03	Stealth Falcon	CVE-2025-33053 used as a zero-day (Check Point via The Record, 2025-06-10)	A major defense organization in Türkiye	Check Point, primary not opened	Reported	[50]
2025-06-10	(CVE)	CVE-2025-33053 added to KEV and published by NVD	Windows users	CISA and NVD	Confirmed	[51] [56]
2026-01-08	(no actor named)	CISA closes ED 19-01	US federal agencies	CISA	Confirmed	[44]
2026-07-31	Several groups	Live ATT&CK pages for G0050, G0040, G0012, G0038, G0121, G0134, G1002, G1019 and G1041 show v20.0 content last modified on this date; G0070 shows version 1.4, last modified 2 September 2026	Not applicable	MITRE	Confirmed	[20] [32]

Reading the record by era. Before 2014 the public record is mostly "since" dates assigned later. From 2014 to 2019 the large vendor reports land (Darkhotel, APT32, Stealth Falcon, Dark Caracal, Sea Turtle), and the US government responds to DNS tampering with an emergency directive. From 2020 on, platform takedowns (Meta) join the record, and the targeting shifts toward students, mobile users and, for SideWinder, maritime and nuclear sectors (**Inferred** from the table).

Part 8: tooling

The library holds a generated page for every Enterprise software object below; the counts and names come from the library's group data, which matched the live group pages (**Confirmed** [32]).

Group	ATT&CK software (Enterprise)
APT32	Arp S0099, Cobalt Strike S0154, Denis S0354, Goopy S0477, ipconfig S0100, Kerrdown S0585, KOMPROGO S0156, Mimikatz S0002, Net S0039, netsh S0108, OSX_OCEANLOTUS.D S0352, PHOREAL S0158, RotaJakiro S1078, SOUNDBITE S0157, WINDSHIELD S0155

Group	ATT&CK software (Enterprise)
Patchwork	AutoIt backdoor S0129, BackConfig S0475, BADNEWS S0128, NDiskMonitor S0272, PowerSploit S0194, QuasarRAT S0262, TINYTYPHON S0131, Unknown Logger S0130; Mobile-only VajraSpy (S9006, no library page)
Sidewinder	Koadic S0250
Confucius	WarzoneRAT S0670
BITTER	ZxxZ S1013
Transparent Tribe	Crimson S0115, DarkComet S0334, njRAT S0385, ObliqueRAT S0644, Peppy S0643
SideCopy	Action RAT S1028, AuTo Stealer S1029
Dark Caracal	Bandook S0234, CrossRAT S0235, FinFisher S0182; Mobile-only Pallas (S0399, no library page)
Volatile Cedar	Caterpillar WebShell S0572, Explosive S0569
POLONIUM	CreepyDrive S1023, CreepySnail S1024
Molerats	DropBook S0547, DustySky S0062, MoleNet S0553, PoisonIvy S0012, SharpStage S0546, Spark S0543
WIRTE	AshTag S9031, Empire S0363, Ferocious S0679, Havoc S1229, IronWind S9029, LitePower S0680, Rclone S1040, SameCoin S9030
APT-C-23	Micropsia S0339
Sea Turtle	SnappyTCP S1163
MoustachedBouncer	Disco S1088, NightClub S1090, SharpDisco S1089
PROMETHIUM	StrongPity S0491, Truvasys S0178
NEODYMIUM	Wingbird S0176
The White Company	NETWIRE S0198, Revenge RAT S0379
Gorgon Group	NanoCore S0336, QuasarRAT S0262, Remcos S0332, njRAT S0385
Higaisa	PlugX S0013, certutil S0160, gh0st RAT S0032
Machete	Machete S0409
Metador	Mafalda S1060, metaMain S1059
Darkhotel, Stealth Falcon, BlackOasis, Winter Vivern	None in ATT&CK

Three patterns stand out. Commodity remote access tools (njRAT, DarkComet, QuasarRAT, NETWIRE, Revenge RAT, PoisonIvy) recur across South Asian and Gaza groups, so a hit on one is a weak attribution signal (**Inferred**). Commercial surveillance software appears directly (FinFisher with Dark Caracal; BlackOasis as a Gamma Group customer), which means the "state actor" is sometimes a buyer of tooling (**Confirmed** [15] [40]; the conclusion is **Inferred**). And the two infrastructure-level actors carry small, purpose-built toolsets: one

implant for Sea Turtle, three for MoustachedBouncer (**Confirmed** [32]).

Part 9: exploited vulnerabilities

Every CVE that the opened primaries and live ATT&CK pages name for this group set, 22 in all, is in the CISA KEV catalog (catalog version 2026.11.03, 1,767 entries) (**Confirmed** [56]). Six of them carry KEV's "Known" flag for use in ransomware campaigns.

CVE	Named for	KEV ransomware use "Known"	Source
CVE-2025-33053	Stealth Falcon (Reported)	No	[50] [51] [56]
CVE-2021-1732	MoustachedBouncer (Disco plugin), BITTER	Yes	[48] [6] [56]
CVE-2017-11882	Sidewinder, BITTER, Patchwork	Yes	[4] [6] [3] [53] [56]
CVE-2018-0802	BITTER	Yes	[6] [56]
CVE-2018-0798	BITTER	No	[6] [56]
CVE-2012-0158	BITTER, Transparent Tribe	Yes	[6] [7] [56]
CVE-2017-0199	Patchwork	Yes	[3] [56]
CVE-2017-8570	Patchwork	No	[3] [56]
CVE-2012-1856	Patchwork	No	[3] [56]
CVE-2014-4114	Patchwork	No	[3] [56]
CVE-2015-1641	Patchwork	No	[3] [56]
CVE-2010-3333	Transparent Tribe	No	[7] [56]
CVE-2020-0674	Sidewinder	No	[4] [56]
CVE-2015-8651	Darkhotel	No	[26] [56]
CVE-2009-1151	Sea Turtle	No	[42] [56]
CVE-2014-6271	Sea Turtle	No	[42] [56]
CVE-2017-3881	Sea Turtle	No	[42] [56]
CVE-2017-6736	Sea Turtle	No	[42] [56]
CVE-2017-12617	Sea Turtle	No	[42] [56]
CVE-2018-0296	Sea Turtle	No	[42] [56]
CVE-2018-7600	Sea Turtle	Yes	[42] [56]
CVE-2025-27920	Marbled Dust (Sea Turtle overlap)	No	[46] [56]

Microsoft's report for Marbled Dust (Part 5) names CVE-2025-27920 in Output Messenger; CISA added it to KEV on 19 May 2025 with CWE-22 (**Confirmed** [46] [56]).

What this means for vulnerability management:

1. **Office build currency is the control that matters for this tier.** Nine of the 22 are Microsoft Office or its MSCOMCTL control (KEV lists CVE-2014-4114 under Windows), and most date from 2010 to 2018 (**Confirmed** [56]). All nine Office CVEs entered KEV between 3 November 2021 and 3 March 2022, in its initial backfill, and 19 of the 22 were added between 3 November 2021 and 25 March 2022; the exceptions are CVE-2015-8651 (25 May 2022), CVE-2025-27920 (19 May 2025) and CVE-2025-33053 (10 June 2025) (**Confirmed** [56]). A remediation SLA keyed only to new KEV additions never raises them (**Inferred**).
2. **Seven are internet-edge and server flaws** that Talos listed for Sea Turtle, with Talos's own warning that the list is likely incomplete (**Confirmed** [42]).
3. **Ransomware and state exploitation overlap on the same flaws.** Six of the 22 carry KEV's "Known" ransomware flag (**Confirmed** [56]), so one patch closes both doors (**Inferred**).
4. **The highest-impact routes have no CVE.** Registrar and DNS account compromise (Sea Turtle) and ISP-level injection (MoustachedBouncer) need non-CVE exposure items: registrar multi-factor authentication and registry lock, Certificate Transparency monitoring and encrypted egress from untrusted networks (**Inferred** from [42] [44] [48]).

Part 10: targets by sector and geography, and the healthcare view

By sector and geography

Sector	Groups and evidence	Geography	Source
Government, foreign ministries, diplomats	Patchwork, Sidewinder, BITTER, Transparent Tribe, SideCopy, Sea Turtle, MoustachedBouncer, Winter Vivern, UNC1151, The White Company, Machete, Higaisa, APT32	South Asia; MENA; Belarus; Europe; Latin America; Southeast Asia	[3] [4] [6] [7] [9] [20] [24] [25] [27] [28] [29] [42] [47]
Military and defense industry	Sidewinder, Transparent Tribe, Patchwork (Pakistan's Ministry of Defense, 2021), POLONIUM, Stealth Falcon (defense organization in Türkiye, 2025, Reported), Dark Caracal	South Asia; Israel; Türkiye	[4] [7] [14] [36] [40] [50]
Civil society: journalists, activists, dissidents, diaspora	Stealth Falcon, Dark Caracal, APT32, BlackOasis	UAE and London; 21+ countries; Vietnam and abroad	[11] [15] [35] [40] [49]
Telecom, ISPs, IT and hosting providers, registrars	Sea Turtle, Metador, SideWinder, Marbled Dust	MENA; the Netherlands; Africa	[30] [42] [45] [46] [53]
Energy, nuclear, maritime and logistics	BITTER (energy), Sea Turtle (energy bodies), SideWinder (nuclear, maritime)	Pakistan, China, Bangladesh, Saudi Arabia; MENA; South Asia	[6] [42] [54]
Academia and research	Transparent Tribe (students, C0011), Patchwork (molecular medicine faculty), Metador (universities), SideWinder (universities)	India; Pakistan; Middle East and Africa	[8] [36] [30] [53]
Business	APT32 (manufacturing, hospitality, banking, media and more), Darkhotel (traveling executives)	Vietnam, Germany, the Philippines, China, the US, Australia; East Asia	[34] [35] [52]

The healthcare view

This tier does not target healthcare as a mission. None of the in-scope ATT&CK descriptions names healthcare (**Inferred** from [1]). What the vendor record shows is healthcare as collateral or opportunity inside wider regional and political targeting (**Inferred** [35] [36] [40]):

- **APT32:** Meta lists hospitals among the businesses APT32 targeted, alongside IT, hospitality, agriculture, commodities, retail, automotive and mobile services (**Confirmed** [35]). Mandiant's 2017 sector list does not name healthcare (**Confirmed** [34]).
- **Patchwork:** the late-2021 campaign reached faculty researching molecular medicine and biological sciences for the first time, as part of a campaign against Pakistan's defense and research establishment (**Confirmed** [36]).
- **Dark Caracal:** medical professionals were among the victims whose devices were compromised (**Confirmed** [40]).
- **Transparent Tribe:** health themes appeared as lures, not targets: a trojanized imitation of the Aarogya Setu COVID-19 tracking app in 2020 and a mental health survey document in 2022 (**Confirmed** [37] [38]).

For a hospital or health system in the US or Europe, the practical exposure from this tier is staff mobile devices (trojanized messaging and health apps), clinicians and researchers who travel, and research collaborations with South and Southeast Asian institutions (**Inferred** [53] and the target lists above). The 2020 joint CISA and NCSC advisory AA20-126A shows the general pattern of state actors going after COVID-19 response bodies (healthcare, pharmaceutical companies, universities, medical research and local government) through password spraying and VPN and Citrix exploitation (`T1110.003`, `T1190`), but it names no group or state, so it is context, not attribution (**Confirmed** [55]).

The sustained, named healthcare harm in the government record comes from the larger programs, summarized in Part 11: North Korean ransomware against hospitals, Iranian access brokering, and espionage against vaccine research (**Inferred** from Part 11). The 2024 Change Healthcare ransomware attack has its own study, Change Healthcare 2024.

Part 11: the wider record, the five largest programs in brief

The groups above are one tier of a larger landscape. The five largest state programs each have a companion study in this series; this Part gives the dated public record for each in short, so the tiers can be compared, with particular attention to the platforms public reporting links to the United States and to the healthcare record. Events with their own case studies are linked, not repeated.

Platforms linked in reporting to the United States

ATT&CK names no sponsor for the Equation group (`G0020`); it describes zero-day exploit use and the ability to overwrite hard-disk firmware (**Confirmed** [31]). The vendor record is detailed about the tools and silent about the state:

- **Equation (Kaspersky, 16 February 2015).** Kaspersky GReAT named the Equation group, linked it to the Stuxnet and Flame groups, and named no state. Its tools are EquationLaser, EquationDrug, DoubleFantasy, TripleFantasy, Fanny, GrayFish and GROK, and one module can reprogram firmware on more than a dozen hard-drive brands. Kaspersky estimated thousands, possibly tens of thousands, of victims in government, telecoms, aerospace, energy, nuclear and finance (**Confirmed** [57]).
- **Fanny and Stuxnet (17 February 2015).** The Fanny worm used two zero-days, the LNK flaw CVE-2010-2568 and MS09-025, that Stuxnet added later (June 2009 and March 2010), and Kaspersky concluded Equation had them years earlier. Sinkhole data showed more than 11,200 IP addresses, 59.36% in Pakistan (**Confirmed** [58]). Stuxnet (`S0603`) has its own study, Stuxnet.
- **Flame (28 May 2012).** About 20 MB of attack toolkit with victims in Iran, Lebanon, Syria and Israel; Kaspersky said a nation state probably sponsored it but that nothing ties Flame to a specific state (**Confirmed** [59]; ATT&CK object `S0143`).
- **Duqu 2.0 (10 June 2015).** Kaspersky found a successor to the 2011 Duqu inside its own network, with some 2014 to 2015 infections tied to venues of the P5+1 Iran nuclear talks; no state named (**Confirmed** [60]; Duqu is `S0038`).
- **Regin (24 November 2014).** 27 victims in 14 countries, among them Belgium, Germany, Iran, Russia and Syria, and a log of GSM base-station controller commands against 136 cells in April and May 2008. Kaspersky called nation-state support likely, warned the metadata could be a false flag, and named no state (**Confirmed** [61]; `S0019`).

- **The Lamberts (11 April 2017).** Kaspersky's name for the actor Symantec calls Longhorn, with colour-coded tools (Black, Blue, Green, White, Pink and Gray Lambert) in use since at least 2008 and first exposed in 2014 through Black Lambert and CVE-2014-4148; no state named (**Confirmed** [63]; Green Lambert is `S0690`).
- **Vault 7 (7 March 2017).** WikiLeaks' "Year Zero" release claims to be 8,761 documents from the CIA's Center for Cyber Intelligence and names the Engineering Development Group and UMBRAGE. This is the leak publisher's claim; authenticity was never officially confirmed (**Reported** [62]).
- **The Shadow Brokers dump and WannaCry (April and May 2017).** Microsoft's security response center found that EternalBlue, EternalRomance, EternalChampion and EternalSynergy were already fixed by MS17-010, with other exploits covered by MS08-067, MS09-050, MS10-061 and MS14-068 (**Confirmed** [64]). After WannaCry, Microsoft stated that the exploits were stolen from the NSA, noted the patch had shipped on 14 March 2017, and called for a Digital Geneva Convention: a vendor statement, not a US government admission (**Confirmed** [65]).
- **Operation Triangulation (1 June 2023).** Kaspersky disclosed a zero-click iMessage chain on iOS found on its staff phones, with traces back to 2019, and made no state attribution; a 2025 update notes that MITRE added an ATT&CK campaign page (**Confirmed** [66]). That campaign is not in the library's data, so it is not linked. The Russian FSB statement blaming US intelligence could not be opened and is not relied on.
- **Northwestern Polytechnical University (disclosed 22 June 2022; report 5 September 2022).** China's CVERC, working with 360, attributes the intrusion to NSA TAO, saying TAO used more than 40 NSA-specific tools (it names FoxAcid, SecondDate and NOPEN) and that more than 1,100 attack chains were found. This is a rival government's accusation, and the attribution is unverified (**Reported** [67]).

Taken together: no US government body has attributed any of these platforms to itself. The sponsor link rests on leaks (the Shadow Brokers and Vault 7), Microsoft's 2017 statement about NSA-origin exploits, and China's CVERC accusation (a Russian FSB statement on Triangulation exists but could not be opened), while the vendors describe the tools without naming a state (**Inferred** [57] to [67]). That is the reverse of the big-four record below, where Western governments publish named-unit attributions. A defender's lesson from this set is about capability, not attribution: firmware implants, GSM network access and zero-click mobile chains were all in the record by 2015 to 2023.

China

China's public record moved from vendor exposure to multinational attribution and sanctions; its companion study is China-Attributed Threat Actors: The Public Record. Mandiant attributed APT1 (G0006) to PLA Unit 61398 of the General Staff 3rd Department, 2nd Bureau, with 141 victims and economic espionage since 2006 (**Confirmed** [68], 19 February 2013). On 20 December 2018 the UK's NCSC held APT10 (G0045, Cloud Hopper) responsible "with the highest level of probability" and judged the MSS responsible, for a campaign against managed service providers and their clients since at least 2016 (**Confirmed** [69]). Microsoft attributed the Exchange zero-day exploitation (CVE-2021-26855, -26857, -26858, -27065) to HAFNIUM (G0125), a state-sponsored group operating from China whose targets included infectious disease researchers (**Confirmed** [70], 2 March 2021); that event's home is Operation Exchange Marauder. The White House attributed that exploitation to MSS-affiliated actors with a high degree of confidence on 19 July 2021, joined by the EU, UK and NATO, and named contract hackers doing ransomware for profit and four MSS hackers charged over a campaign that included healthcare (**Confirmed** [71]). The same day, AA21-200A tied the indicted APT40 (G0065) actors to the MSS Hainan State Security Department and its front company Hainan Xiandun, active since at least 2009 (**Confirmed** [72]). In July 2024, eight countries led by Australia's ACSC assessed that APT40 acts for the MSS and turns public proofs of concept into working exploits within hours or days, including Log4Shell CVE-2021-44228 (**Confirmed** [73]; see Log4Shell).

Volt Typhoon (G1017) marked a shift from theft to preparation. Microsoft reported on 24 May 2023 a China-based state-sponsored actor active since mid-2021, with targets in Guam and elsewhere in the US, building, with moderate confidence, the ability to disrupt US to Asia communications in a crisis, with initial access through Fortinet FortiGuard devices (**Confirmed** [74]). The same-day advisory AA23-144A described its living-off-the-land tradecraft (**Confirmed** [75]), and on 7 February 2024 AA24-038A assessed with high confidence that it was pre-positioning on IT networks in communications, energy, transport and water to disrupt operational technology in a crisis, with some footholds held for at least five years (**Confirmed** [76]). Salt Typhoon (G1045) followed: on 27 August 2025, 23 agencies from 13 countries described actors targeting backbone, provider-edge and customer-edge routers of major telecoms since at least 2021, and named three Chinese companies (Sichuan Juxinhe, Beijing Huanyu Tianqiong and Sichuan Zhixin Ruijie) that supply the PLA and MSS (**Confirmed** [77]). On 25 March 2024 the UK said APT31 (G0128, ATT&CK's ZIRCONIUM) ran 2021 reconnaissance against parliamentarians and that a China state-affiliated actor likely compromised the Electoral Commission between late 2021 and October 2022, sanctioning the MSS front Wuhan Xiaoruizhi and two APT31 members; the release uses both "almost certain" and "highly likely" in different places (**Confirmed** [78]). On 3 January 2025 Treasury sanctioned Integrity Technology Group of Beijing for supporting Flax Typhoon, active since at least 2021 (**Confirmed** [79]). For healthcare, the FBI and CISA warned on 13 May 2020 of likely PRC targeting and compromise of COVID-19 research, healthcare and pharmaceutical organizations (**Confirmed** [80]). The fullest single-group record is in APT41: The Public Record.

Russia

The companion study is Russia-Attributed Threat Actors: The Public Record. On 4 October 2018 the UK's NCSC said the GRU was almost certainly responsible for a campaign of attacks, naming aliases including APT28 (G0007) and Sandworm, and operations against WADA medical files (August 2016), the DNC (2016), Bad Rabbit (S0606, October 2017) and attempts on the OPCW (2018) (**Confirmed** [81]). On 16 July 2020 NCSC and Canada's CSE assessed that APT29 (G0016) is almost certainly part of the Russian intelligence services and had targeted COVID-19 vaccine development in Canada, the US and the UK with WellMess (S0514) and WellMail (S0515), with NSA agreement (**Confirmed** [82]). The White House named the SVR as the perpetrator of the SolarWinds Orion campaign on 15 April 2021; that event's home is SolarWinds SUNBURST (**Confirmed** [84]). AA21-116A covered SVR tradecraft (**Confirmed** [83]); AA23-347A reported SVR exploitation of JetBrains TeamCity CVE-2023-42793 from late September 2023, compromising a few dozen companies opportunistically while possibly preparing supply-chain access (**Confirmed** [85]); and Microsoft reported that Midnight Blizzard password-sprayed a legacy test tenant from late November 2023 and reached senior leadership email, detected on 12 January 2024 (**Confirmed** [89]). On 9 May 2023 Five Eyes agencies attributed the Snake implant (developed as Uroburos, S0022, from late 2003) to a unit of FSB Center 16 that also runs Turla (G0010), with infrastructure in more than 50 countries (**Confirmed** [86]). AA25-141A (21 May 2025) attributed to GRU Unit 26165 (APT28) a campaign since 2022 against logistics and technology firms that move aid to Ukraine and against IP cameras, about 81% of them in Ukraine (**Confirmed** [87]). AA24-249A (5 September 2024) attributed operations to GRU Unit 29155, overlapping Cadet Blizzard and Ember Bear (G1003), including WhisperGate (S0689) from 13 January 2022, scanning across 26 NATO members and healthcare among the targeted sectors (**Confirmed** [88]). Sandworm's record is in Sandworm Operations.

North Korea

North Korea's record is the one most tied to healthcare harm. WannaCry (S0366) hit NHS England on 12 May 2017: the UK National Audit Office found at least 81 of 236 trusts and 595 GP practices affected, with 6,912 cancelled appointments recorded and an estimated 19,000 or more, and unpatched or unsupported Windows as the common weakness; the report names no perpetrator (**Confirmed** [91]). On 19 December 2017 the UK said the NCSC assessed it highly likely that Lazarus Group (G0032) was behind WannaCry, which hit about 300,000 computers in 150 countries including 48 NHS trusts (**Confirmed** [90]). On 13 September 2019 Treasury sanctioned Lazarus Group, Bluenoroff and Andariel (G0138) as controlled by the RGB, citing Sony Pictures (2014), the theft of about $80M from Bangladesh Bank (2016) and WannaCry, including over $112M in NHS cost (**Confirmed** [92]). The NHS harm was collateral damage, not targeting (**Inferred** [91] [92]).

The money record runs alongside: the BeagleBoyz, an RGB element overlapping APT38 (G0082) to varying degrees, abused SWIFT from 2015 and ran FASTCash ATM cash-outs from 2016 against banks in 38 nations, with nearly $2B attempted by public estimate (**Confirmed** [93]); TraderTraitor trojanized crypto applications from at least 2020 and links to AppleJeus (S0584) (**Confirmed** [95]); and the FBI attributed the theft of about $1.5B from Bybit, on or about 21 February 2025, to TraderTraitor (**Confirmed** [99]). Kimsuky (G0094), active since about 2012, is most likely tasked by the regime with global intelligence gathering (**Confirmed** [94]).

Then the targeting of healthcare became direct. The FBI assessed that North Korean state-sponsored actors had used Maui ransomware against US healthcare and public health organizations since May 2021, encrypting electronic health record, diagnostics, imaging and intranet servers (**Confirmed** [96]). A February 2023 joint advisory from US agencies, HHS and South Korea's NIS and DSA described state-sponsored

ransomware (Maui, H0lyGh0st, sometimes posing as REvil) against US and South Korean healthcare, with entry through Log4Shell CVE-2021-44228, SonicWall CVE-2021-20038 and TerraMaster CVE-2022-24990, and a trojanized X-Popup messenger used by Korean hospitals (**Confirmed** [97]). In July 2024, agencies including NCSC-UK and the ROK NIS placed Andariel in the RGB 3rd Bureau and said it funds espionage against defense, aerospace, nuclear and engineering targets with ransomware against US healthcare, sometimes hitting the same victim for both (**Confirmed** [98]).

Iran

Iran's record mixes state espionage, destructive attacks and access brokering into the ransomware economy. Treasury sanctioned two Iran-based men in November 2018 for converting SamSam ransom bitcoin into rial; SamSam had more than 200 victims including hospitals, and Treasury does not state state sponsorship (**Confirmed** [106]). Treasury sanctioned APT39 (G0087), 45 individuals and the front company Rana Intelligence Computing for MOIS work against dissidents, journalists and the travel sector on 17 September 2020 (**Confirmed** [102]). In June 2021 an Iranian government-sponsored APT exploiting Fortinet and ProxyShell flaws reached the environmental control networks of a US children's hospital (**Confirmed** [100]). AA22-055A described MuddyWater (G0069) as a subordinate element of the MOIS, active since about 2018, with PowGoop (S1046), Small Sieve (S1035), Canopy (ATT&CK's STARWHALE, S1037), Mori (S1047) and POWERSTATS (S0223) (**Confirmed** [101] [32]). Iranian state actors calling themselves HomeLand Justice hit the Government of Albania with ransomware and wipers in July 2022 after about 14 months of access, then struck again in September 2022 (**Confirmed** [107]). IRGC-affiliated CyberAv3ngers compromised at least 75 Unitronics PLCs and HMIs, at least 34 in US water, between 22 November 2023 and January 2024, with healthcare among the affected sectors; the advisory maps only Enterprise ATT&CK, and CyberAv3ngers (G1027) has no library group page (**Confirmed** [103] [32]). The FBI assesses that Pioneer Kitten (ATT&CK's Fox Kitten, G0117) is Iranian state-sponsored and sells access to NoEscape, RansomHouse and ALPHV affiliates, ransomware work it says is likely not sanctioned by the state, with US healthcare facilities among the victims (**Confirmed** [104]). Since October 2023 Iranian actors have used password spraying and MFA push bombing against healthcare and other sectors, and some of that activity may come from third parties who bought the access (**Confirmed** [105]). Google TAG reported that APT42 (G1044), associated with the IRGC, phished people tied to the Biden and Trump campaigns in May and June 2024 (**Confirmed** [108]). On 30 June 2025 a joint fact sheet warned of possible Iranian activity against US critical infrastructure and firms tied to Israel while saying no coordinated campaign had been observed (**Confirmed** [109]).

How state-actor healthcare harm takes shape

In the government record, state-actor harm to healthcare takes three forms (**Inferred** [96] to [98], [100], [104], [105], [82], [70]):

1. **Ransomware for revenue:** North Korea's Maui and H0lyGh0st operations and Andariel's ransomware-funded espionage (AA22-187A, AA23-040A, AA24-207A).
2. **Access brokering and partnering with ransomware groups:** Iranian actors (AA21-321A, AA24-241A, AA24-290A).
3. **Espionage against vaccine and disease research:** APT29 in 2020 and HAFNIUM in 2021.

WannaCry's harm to the NHS was collateral damage, not targeting. The other-states tier in Parts 1 to 10 fits none of the three forms as a pattern; its healthcare contact is incidental (**Inferred**).

Part 12: the government and platform record for this tier

The big five draw joint advisories, indictments and sanctions. This tier draws far fewer, and they are mostly indirect:

- **CISA Emergency Directive 19-01 (22 January 2019, closed 8 January 2026)** is the one US government action in this record aimed at the DNS-tampering method Sea Turtle used, though it names no actor (**Confirmed** [44]; the link to Sea Turtle is **Inferred**).
- **Meta's takedowns** are the most direct public actions against named groups: APT32 and its tie to CyberOne Group (10 December 2020), and Bitter APT and APT36 (4 August 2022) (**Confirmed** [35] [39]).
- **The Indian Army's warning** about fake Aarogya Setu apps (late April 2020) is known only through Indian press reports cited by Kaspersky (**Reported** [37]).
- **CISA's KEV listing of CVE-2025-33053** on 10 June 2025, the day it was disclosed, is the government trace of Stealth Falcon's reported zero-day (**Confirmed** [56]; the Stealth Falcon link is **Reported** [50]).

No indictment, sanctions designation or joint multinational attribution names any of the 20 bound groups in the verified record (**Inferred** from the sources opened). The contrast is itself a finding: defenders facing this tier work from vendor and platform reporting, not from government advisories with technique tables.

Part 13: the defender view

What a document-and-lure tier means in practice

The technique counts describe a familiar chain: a spearphishing attachment or link, a user opening a file, an old Office or browser exploit or a macro, then script interpreters (PowerShell and Visual Basic, eight groups each) and run-key persistence (eight groups) (**Confirmed** [32]; the reading is **Inferred**). The good news for defenders is that the controls already exist and are well understood. The bad news is that they must work everywhere, including on the laptops of travelers and the phones of staff.

Hardening and hunting leads, by source

For DNS and registrar compromise (Talos [42] [43] and CISA ED 19-01 [44]):

- Registry lock with out-of-band confirmation, and multi-factor authentication on every DNS management and registrar account (**Confirmed** [42] [43] [44]).
- Audit public DNS records, change DNS account passwords, and monitor Certificate Transparency logs for certificates you did not request (**Confirmed** [44]).
- DNSSEC-sign domains and validate in recursive resolvers; restrict IMAP; monitor passive DNS for abnormal record changes (**Confirmed** [43]).

For hosting and IT service providers (Hunt & Hackett [45]): Sea Turtle's 2021 to 2023 access came through compromised cPanel accounts and SSH, and the actor cleared logs and command history (**Confirmed** [45]). Keep centralized logs outside the host, put multi-factor authentication on hosting control panels, and reduce internet-exposed SSH (**Inferred** from [45]).

For in-path injection (ESET [48]): an end-to-end encrypted VPN tunnel, ideally out-of-band rather than from the endpoint, providing connectivity from a trusted network, which ESET recommends for foreign

organizations in Belarus; treat any Windows Update prompt that does not come from Microsoft's update channel as hostile, and watch for SMTP command and control from workstations (**Confirmed** for the VPN advice [48]; the detection points are **Inferred** from [48]).

For dissident and journalist surveillance (Citizen Lab [49], Lookout and EFF [40] [41]): block macros in files from the internet and alert on PowerShell launched from Office (`T1059.001` following `T1204.002`); allow mobile apps from official stores only and block sideloading (**Inferred** from [40] [49]).

For travelers (Kaspersky [52]): use a VPN before anything else on hotel and guest Wi-Fi, and alert on binaries signed with weak-key or revoked certificates (`T1553.002`) (**Inferred** from [52]).

Where the library's countermeasure data is thin

`T1584.002` DNS Server, `T1588.004` Digital Certificates, `T1553.002` Code Signing, `T1659` Content Injection, `T1685.006` Clear Linux or Mac System Logs and `T1690` Prevent Command History Logging have no D3FEND mapping in the library's data (**Confirmed** [32]). Multi-factor Authentication (`D3-MFA`), Software Update (`D3-SU`) and Application Configuration Hardening (`D3-ACH`) have library pages, but the library maps none of them to `T1078`, `T1203`, `T1190`, `T1566.001` or `T1566.002`, so this study cites them as guidance from the sources above, not as library-mapped countermeasures (**Confirmed** [32]). And D3FEND's own Email Filtering page lists `T1566.001` and `T1566.002` as inferred, experimental relationships rather than confirmed mappings (**Confirmed** [33]).

Overstated and understated

- **Overstated: the reach of this tier into Western healthcare.** Nothing in the verified record shows any of these groups running a healthcare campaign; the contacts are incidental (**Inferred**).
- **Overstated: confidence in names.** Many of these groups carry "suspected" or no state at all, and several mimic each other's tradecraft. Reports that present them as settled national programs go further than the sources (**Inferred** from Part 1).
- **Understated: DNS and registrar risk.** Sea Turtle's method compromises the trust layer under everything else, carries no CVE, and was serious enough for a US emergency directive that stayed open for seven years (**Inferred** from [42] [44]).
- **Understated: old Office flaws.** They are routinely treated as solved, yet Kaspersky still recorded CVE-2017-11882 in 2024 operations (**Confirmed** [53]).

Part 14: MITRE mappings

This part binds the verified record to MITRE ATT&CK v20.0, CAPEC, CWE and D3FEND for 20 groups: the 14 whose ATT&CK text names or suspects a state, a state body, a region (BITTER, "suspected South Asian") or a Hamas affiliation outside the big five, or calls them likely state-sponsored, plus 6 that ATT&CK leaves stateless but that vendor primaries or ATT&CK's own overlap notes place in that tier (Confucius, Molerats, APT-C-23, PROMETHIUM, MoustachedBouncer and BlackOasis; the placement is **Inferred** where no source names a state): `G0050`, `G0121`, `G0040`, `G0142`, `G1002`, `G0134`, `G1008`, `G0012`, `G0038`, `G0070`, `G0123`, `G1005`, `G0021`, `G0090`, `G1028`, `G1041`, `G1019`, `G0056`, `G0063` and `G0089` (**Confirmed** [32]). Every ID was checked against the library's data files or the live ATT&CK page on 3 November 2026; IDs that could not be checked are left out. Tactic names are the v20 names (for example `stealth` and `defense-impairment`), read from the library's `technique_profiles.jsonl`.

Enterprise techniques stated by vendor and government primaries

These are the techniques a dated primary describes for a group in this set. Every technique also appears in the ATT&CK record for the set below; three Stealth Falcon entries (`T1566.001`, `T1566.002`, `T1204.002`, Citizen Lab [49]) and the UNC1151 entry are vendor-stated and absent from that group's ATT&CK record (**Confirmed** [32] [47] [49]).

Technique	Name	Tactic	Dated event or report	Source
`T1584.002`	DNS Server	resource-development	Sea Turtle DNS hijacking through registrars and registries, January 2017 to Q1 2019 (Talos, 2019-04-17)	[42]
`T1557`	Adversary-in-the-Middle	credential-access, collection	Sea Turtle credential harvesting, 2017 to 2019	[42]
`T1588.004`	Digital Certificates	resource-development	Sea Turtle certificates for victim domains, 2017 to 2019	[42]
`T1078`	Valid Accounts	stealth, persistence, privilege-escalation, initial-access	Sea Turtle stolen credentials (2017 to 2019); compromised cPanel accounts (2021 to 2023)	[42] [45]
`T1190`	Exploit Public-Facing Application	initial-access	Sea Turtle's seven named CVEs (2019-04-17); Marbled Dust CVE-2025-27920 (2025-05-12)	[42] [46]
`T1133`	External Remote Services	persistence, initial-access	Sea Turtle SSH access, 2021 to 2023	[45]
`T1685.006`	Clear Linux or Mac System Logs	defense-impairment	Sea Turtle log clearing, 2021 to 2023	[45]
`T1690`	Prevent Command History Logging	defense-impairment	Sea Turtle history clearing, 2021 to 2023	[45]
`T1114.001`	Local Email Collection	collection	Sea Turtle email archive, 2021 to 2023	[45]
`T1074.002`	Remote Data Staging	collection	Sea Turtle archive staged in a public web directory, 2021 to 2023	[45]
`T1560.001`	Archive via Utility	collection	Sea Turtle email archive, 2021 to 2023	[45]

Technique	Name	Tactic	Dated event or report	Source
T1659	Content Injection	initial-access, command-and-control	MoustachedBouncer fake Windows Update via ISP-level injection, since 2020 (ESET, 2023-08-10)	[48]
T1068	Exploitation for Privilege Escalation	privilege-escalation	MoustachedBouncer Disco plugin, CVE-2021-1732 (ESET, 2023-08-10)	[48]
T1059.001	PowerShell	execution	Stealth Falcon backdoor (Citizen Lab, 2016-05-29)	[49]
T1573.001	Symmetric Cryptography	command-and-control	Stealth Falcon RC4-encrypted C2 (2016-05-29)	[49]
T1555.003	Credentials from Web Browsers	credential-access	Stealth Falcon browser theft (2016-05-29)	[49]
T1555.004	Windows Credential Manager	credential-access	Stealth Falcon credential-vault theft (2016-05-29)	[49]
T1566.001	Spearphishing Attachment	initial-access	APT32 macro documents (Mandiant, 2017-05-14); Stealth Falcon (2016-05-29); Patchwork RTF lures (December 2021); SideWinder template-injection documents (Kaspersky, 2024-10-15)	[34] [49] [36] [53]
T1566.002	Spearphishing Link	initial-access	Stealth Falcon tracked links (2016-05-29); UNC1151 credential phishing, 2016 to 2021 (not an ATT&CK group)	[49] [47]
T1204.002	Malicious File	execution	APT32 macro documents (2017-05-14); Stealth Falcon macro documents (2016-05-29)	[34] [49]
T1189	Drive-by Compromise	initial-access	Dark Caracal watering-hole site (2018-01-18); Darkhotel hotel-network delivery (2014-11-10)	[40] [52]
T1566.003	Spearphishing via Service	initial-access	Dark Caracal phishing through WhatsApp and Facebook (2018-01-18)	[40]
T1203	Exploitation for Client Execution	execution	Darkhotel Flash zero-days (2014-11-10); Patchwork Equation Editor exploit (December 2021); SideWinder CVE-2017-11882 (2024-10-15)	[52] [36] [53]
T1553.002	Code Signing	defense-impairment	Darkhotel weak and stolen certificates (2014-11-10)	[52]

Enterprise techniques recorded by ATT&CK for the 20 groups

ATT&CK v20.0 links 154 distinct Enterprise techniques to the 20 groups (**Confirmed** [32]); every technique the library lists for these groups appears on the live group page [2] to [28]. The table is sorted by the number of groups linked, then by ID. Counting each technique once per tactic it serves gives 181 technique-tactic entries: Stealth 39, Discovery 19, Execution 17, Resource Development 16, Command and Control 15, Initial Access 12, Collection 11, Persistence 11, Privilege Escalation 9, Credential Access 8, Lateral Movement 8, Defense Impairment 6, Reconnaissance 6 and Exfiltration 4 (computed from [32]).

Technique	Name	Tactic	Groups (n)	Groups	Source
T1204.002	Malicious File	execution	13	G0012 Darkhotel, G0021 Molerats, G0040 Patchwork, G0050 APT32, G0056 PROMETHIUM, G0070 Dark Caracal, G0089 The White Company, G0090 WIRTE, G0121 Sidewinder, G0134 Transparent Tribe, G0142 Confucius, G1002 BITTER, G1008 SideCopy	[32]
T1566.001	Spearphishing Attachment	initial-access	11	G0012 Darkhotel, G0021 Molerats, G0040 Patchwork, G0050 APT32, G0089 The White Company, G0090 WIRTE, G0121 Sidewinder, G0134 Transparent Tribe, G0142 Confucius, G1002 BITTER, G1008 SideCopy	[32]
T1105	Ingress Tool Transfer	command-and-control	10	G0012 Darkhotel, G0021 Molerats, G0040 Patchwork, G0050 APT32, G0090 WIRTE, G0121 Sidewinder, G0123 Volatile Cedar, G0142 Confucius, G1002 BITTER, G1008 SideCopy	[32]
T1203	Exploitation for Client Execution	execution	9	G0012 Darkhotel, G0040 Patchwork, G0050 APT32, G0089 The White Company, G0121 Sidewinder, G0134 Transparent Tribe, G0142 Confucius, G1002 BITTER, G1041 Sea Turtle	[32]
T1036.005	Match Legitimate Resource Name or Location	stealth	8	G0012 Darkhotel, G0040 Patchwork, G0050 APT32, G0056 PROMETHIUM, G0090 WIRTE, G0121 Sidewinder, G0134 Transparent Tribe, G1008 SideCopy	[32]
T1059.001	PowerShell	execution	8	G0021 Molerats, G0038 Stealth Falcon, G0040 Patchwork, G0050 APT32, G0090 WIRTE, G0121 Sidewinder, G0142 Confucius, G1019 MoustachedBouncer	[32]
T1059.005	Visual Basic	execution	8	G0021 Molerats, G0040 Patchwork, G0050 APT32, G0090 WIRTE, G0121 Sidewinder, G0134 Transparent Tribe, G0142 Confucius, G1008 SideCopy	[32]
T1071.001	Web Protocols	command-and-control	8	G0038 Stealth Falcon, G0050 APT32, G0070 Dark Caracal, G0090 WIRTE, G0121 Sidewinder, G0142 Confucius, G1002 BITTER, G1041 Sea Turtle	[32]
T1547.001	Registry Run Keys / Startup Folder	persistence, privilege-escalation	8	G0012 Darkhotel, G0021 Molerats, G0040 Patchwork, G0050 APT32, G0056 PROMETHIUM, G0070 Dark Caracal, G0121 Sidewinder, G0142 Confucius	[32]

Technique	Name	Tactic	Groups (n)	Groups	Source
T1204.001	Malicious Link	execution	7	G0021 Molerats, G0040 Patchwork, G0050 APT32, G0090 WIRTE, G0121 Sidewinder, G0134 Transparent Tribe, G0142 Confucius	[32]
T1566.002	Spearphishing Link	initial-access	7	G0021 Molerats, G0040 Patchwork, G0050 APT32, G0090 WIRTE, G0121 Sidewinder, G0134 Transparent Tribe, G0142 Confucius	[32]
T1027.013	Encrypted/Encoded File	stealth	6	G0012 Darkhotel, G0050 APT32, G0070 Dark Caracal, G0121 Sidewinder, G0134 Transparent Tribe, G1002 BITTER	[32]
T1053.005	Scheduled Task	execution, persistence, privilege-escalation	6	G0021 Molerats, G0038 Stealth Falcon, G0040 Patchwork, G0050 APT32, G0142 Confucius, G1002 BITTER	[32]
T1082	System Information Discovery	discovery	6	G0012 Darkhotel, G0038 Stealth Falcon, G0040 Patchwork, G0050 APT32, G0121 Sidewinder, G1008 SideCopy	[32]
T1083	File and Directory Discovery	discovery	6	G0012 Darkhotel, G0040 Patchwork, G0050 APT32, G0070 Dark Caracal, G0121 Sidewinder, G0142 Confucius	[32]
T1189	Drive-by Compromise	initial-access	6	G0012 Darkhotel, G0040 Patchwork, G0050 APT32, G0056 PROMETHIUM, G0070 Dark Caracal, G0134 Transparent Tribe	[32]
T1588.002	Tool	resource-development	6	G0040 Patchwork, G0050 APT32, G0090 WIRTE, G1002 BITTER, G1005 POLONIUM, G1041 Sea Turtle	[32]
T1016	System Network Configuration Discovery	discovery	5	G0012 Darkhotel, G0038 Stealth Falcon, G0050 APT32, G0121 Sidewinder, G1008 SideCopy	[32]
T1059.003	Windows Command Shell	execution	5	G0012 Darkhotel, G0040 Patchwork, G0050 APT32, G0070 Dark Caracal, G0090 WIRTE	[32]
T1518.001	Security Software Discovery	discovery	5	G0012 Darkhotel, G0040 Patchwork, G0089 The White Company, G0121 Sidewinder, G1008 SideCopy	[32]
T1574.001	DLL	stealth, execution	5	G0040 Patchwork, G0050 APT32, G0090 WIRTE, G0121 Sidewinder, G1008 SideCopy	[32]
T1583.001	Domains	resource-development	5	G0050 APT32, G0090 WIRTE, G0134 Transparent Tribe, G1002 BITTER, G1041 Sea Turtle	[32]
T1027.002	Software Packing	stealth	4	G0040 Patchwork, G0070 Dark Caracal, G0089 The White Company, G1019 MoustachedBouncer	[32]

Technique	Name	Tactic	Groups (n)	Groups	Source
T1027.010	Command Obfuscation	stealth	4	G0040 Patchwork, G0050 APT32, G0090 WIRTE, G0121 Sidewinder	[32]
T1033	System Owner/User Discovery	discovery	4	G0038 Stealth Falcon, G0040 Patchwork, G0050 APT32, G0121 Sidewinder	[32]
T1041	Exfiltration Over C2 Channel	exfiltration	4	G0038 Stealth Falcon, G0050 APT32, G0090 WIRTE, G0142 Confucius	[32]
T1057	Process Discovery	discovery	4	G0012 Darkhotel, G0021 Molerats, G0038 Stealth Falcon, G0121 Sidewinder	[32]
T1059.007	JavaScript	execution	4	G0021 Molerats, G0050 APT32, G0121 Sidewinder, G1019 MoustachedBouncer	[32]
T1218.005	Mshta	stealth	4	G0050 APT32, G0121 Sidewinder, G0142 Confucius, G1008 SideCopy	[32]
T1553.002	Code Signing	defense-impairment	4	G0012 Darkhotel, G0021 Molerats, G0040 Patchwork, G0056 PROMETHIUM	[32]
T1608.001	Upload Malware	resource-development	4	G0050 APT32, G0090 WIRTE, G1002 BITTER, G1008 SideCopy	[32]
T1005	Data from Local System	collection	3	G0038 Stealth Falcon, G0040 Patchwork, G0070 Dark Caracal	[32]
T1036.004	Masquerade Task or Service	stealth	3	G0050 APT32, G0056 PROMETHIUM, G1002 BITTER	[32]
T1068	Exploitation for Privilege Escalation	privilege-escalation	3	G0050 APT32, G1002 BITTER, G1019 MoustachedBouncer	[32]
T1070.004	File Deletion	stealth	3	G0040 Patchwork, G0050 APT32, G0089 The White Company	[32]
T1074.001	Local Data Staging	collection	3	G0040 Patchwork, G0090 WIRTE, G0121 Sidewinder	[32]
T1078.003	Local Accounts	stealth, persistence, privilege-escalation, initial-access	3	G0050 APT32, G0056 PROMETHIUM, G1041 Sea Turtle	[32]
T1119	Automated Collection	collection	3	G0040 Patchwork, G0121 Sidewinder, G0142 Confucius	[32]
T1124	System Time Discovery	discovery	3	G0012 Darkhotel, G0089 The White Company, G0121 Sidewinder	[32]
T1140	Deobfuscate/Decode Files or Information	stealth	3	G0012 Darkhotel, G0021 Molerats, G0090 WIRTE	[32]
T1505.003	Web Shell	persistence	3	G0050 APT32, G0123 Volatile Cedar, G1041 Sea Turtle	[32]

Technique	Name	Tactic	Groups (n)	Groups	Source
T1555.003	Credentials from Web Browsers	credential-access	3	G0021 Molerats, G0038 Stealth Falcon, G0040 Patchwork	[32]
T1559.002	Dynamic Data Exchange	execution	3	G0040 Patchwork, G0121 Sidewinder, G1002 BITTER	[32]
T1583.006	Web Services	resource-development	3	G0050 APT32, G0142 Confucius, G1005 POLONIUM	[32]
T1598.003	Spearphishing Link	reconnaissance	3	G0040 Patchwork, G0050 APT32, G0121 Sidewinder	[32]
T1012	Query Registry	discovery	2	G0038 Stealth Falcon, G0050 APT32	[32]
T1027.015	Compression	stealth	2	G0021 Molerats, G0090 WIRTE	[32]
T1047	Windows Management Instrumentation	execution	2	G0038 Stealth Falcon, G0050 APT32	[32]
T1056.001	Keylogging	collection, credential-access	2	G0012 Darkhotel, G0050 APT32	[32]
T1059	Command and Scripting Interpreter	execution	2	G0038 Stealth Falcon, G0050 APT32	[32]
T1074.002	Remote Data Staging	collection	2	G1019 MoustachedBouncer, G1041 Sea Turtle	[32]
T1078	Valid Accounts	stealth, persistence, privilege-escalation, initial-access	2	G1005 POLONIUM, G1041 Sea Turtle	[32]
T1090	Proxy	command-and-control	2	G1005 POLONIUM, G1019 MoustachedBouncer	[32]
T1106	Native API	execution	2	G0090 WIRTE, G1008 SideCopy	[32]
T1112	Modify Registry	defense-impairment, persistence	2	G0040 Patchwork, G0050 APT32	[32]
T1113	Screen Capture	collection	2	G0070 Dark Caracal, G1019 MoustachedBouncer	[32]
T1114.001	Local Email Collection	collection	2	G0090 WIRTE, G1041 Sea Turtle	[32]
T1190	Exploit Public-Facing Application	initial-access	2	G0123 Volatile Cedar, G1041 Sea Turtle	[32]
T1199	Trusted Relationship	initial-access	2	G1005 POLONIUM, G1041 Sea Turtle	[32]
T1218.010	Regsvr32	stealth	2	G0050 APT32, G0090 WIRTE	[32]
T1497.001	System Checks	stealth, discovery	2	G0012 Darkhotel, G0090 WIRTE	[32]
T1518	Software Discovery	discovery	2	G0121 Sidewinder, G1008 SideCopy	[32]
T1543.003	Windows Service	persistence, privilege-escalation	2	G0050 APT32, G0056 PROMETHIUM	[32]
T1560	Archive Collected Data	collection	2	G0040 Patchwork, G0050 APT32	[32]

Technique	Name	Tactic	Groups (n)	Groups	Source
T1564.001	Hidden Files and Directories	stealth	2	G0050 APT32, G0134 Transparent Tribe	[32]
T1567.002	Exfiltration to Cloud Storage	exfiltration	2	G0142 Confucius, G1005 POLONIUM	[32]
T1568	Dynamic Resolution	command-and-control	2	G0134 Transparent Tribe, G1002 BITTER	[32]
T1571	Non-Standard Port	command-and-control	2	G0050 APT32, G0090 WIRTE	[32]
T1573.001	Symmetric Cryptography	command-and-control	2	G0012 Darkhotel, G0038 Stealth Falcon	[32]
T1584.001	Domains	resource-development	2	G0134 Transparent Tribe, G1008 SideCopy	[32]
T1587.002	Code Signing Certificates	resource-development	2	G0040 Patchwork, G0056 PROMETHIUM	[32]
T1598.002	Spearphishing Attachment	reconnaissance	2	G0121 Sidewinder, G1008 SideCopy	[32]
T1608.004	Drive-by Target	resource-development	2	G0050 APT32, G0134 Transparent Tribe	[32]
T1680	Local Storage Discovery	discovery	2	G0040 Patchwork, G0142 Confucius	[32]
T1003	OS Credential Dumping	credential-access	1	G0050 APT32	[32]
T1003.001	LSASS Memory	credential-access	1	G0050 APT32	[32]
T1018	Remote System Discovery	discovery	1	G0050 APT32	[32]
T1020	Automated Exfiltration	exfiltration	1	G0121 Sidewinder	[32]
T1021.001	Remote Desktop Protocol	lateral-movement	1	G0040 Patchwork	[32]
T1021.002	SMB/Windows Admin Shares	lateral-movement	1	G0050 APT32	[32]
T1027	Obfuscated Files or Information	stealth	1	G0063 BlackOasis	[32]
T1027.001	Binary Padding	stealth	1	G0040 Patchwork	[32]
T1027.004	Compile After Delivery	stealth	1	G1041 Sea Turtle	[32]
T1027.005	Indicator Removal from Tools	stealth	1	G0040 Patchwork	[32]
T1027.011	Fileless Storage	stealth	1	G0050 APT32	[32]
T1027.016	Junk Code Insertion	stealth	1	G0050 APT32	[32]
T1036	Masquerading	stealth	1	G0050 APT32	[32]

Technique	Name	Tactic	Groups (n)	Groups	Source
T1036.003	Rename Legitimate Utilities	stealth	1	G0050 APT32	[32]
T1046	Network Service Discovery	discovery	1	G0050 APT32	[32]
T1048.003	Exfiltration Over Unencrypted Non-C2 Protocol	exfiltration	1	G0050 APT32	[32]
T1049	System Network Connections Discovery	discovery	1	G0050 APT32	[32]
T1055	Process Injection	stealth, privilege-escalation	1	G0050 APT32	[32]
T1055.012	Process Hollowing	stealth, privilege-escalation	1	G0040 Patchwork	[32]
T1059.004	Unix Shell	execution	1	G1041 Sea Turtle	[32]
T1070.006	Timestomp	stealth	1	G0050 APT32	[32]
T1071.003	Mail Protocols	command-and-control	1	G0050 APT32	[32]
T1072	Software Deployment Tools	execution, lateral-movement	1	G0050 APT32	[32]
T1080	Taint Shared Content	lateral-movement	1	G0012 Darkhotel	[32]
T1087.001	Local Account	discovery	1	G0050 APT32	[32]
T1091	Replication Through Removable Media	lateral-movement, initial-access	1	G0012 Darkhotel	[32]
T1095	Non-Application Layer Protocol	command-and-control	1	G1002 BITTER	[32]
T1102	Web Service	command-and-control	1	G0050 APT32	[32]
T1102.001	Dead Drop Resolver	command-and-control	1	G0040 Patchwork	[32]
T1102.002	Bidirectional Communication	command-and-control	1	G1005 POLONIUM	[32]
T1132.001	Standard Encoding	command-and-control	1	G0040 Patchwork	[32]
T1133	External Remote Services	persistence, initial-access	1	G1041 Sea Turtle	[32]
T1135	Network Share Discovery	discovery	1	G0050 APT32	[32]
T1137	Office Application Startup	persistence	1	G0050 APT32	[32]
T1197	BITS Jobs	stealth, persistence, execution	1	G0040 Patchwork	[32]
T1205.001	Port Knocking	stealth, persistence, command-and-control	1	G0056 PROMETHIUM	[32]

Technique	Name	Tactic	Groups (n)	Groups	Source
T1213.006	Databases	collection	1	G1041 Sea Turtle	[32]
T1216.001	PubPrn	stealth	1	G0050 APT32	[32]
T1218.001	Compiled HTML File	stealth	1	G0070 Dark Caracal	[32]
T1218.007	Msiexec	stealth	1	G0021 Molerats	[32]
T1218.011	Rundll32	stealth	1	G0050 APT32	[32]
T1221	Template Injection	stealth	1	G0142 Confucius	[32]
T1222.002	Linux and Mac Permissions	defense-impairment	1	G0050 APT32	[32]
T1497	Virtualization/Sandbox Evasion	stealth, discovery	1	G0012 Darkhotel	[32]
T1497.002	User Activity Based Checks	stealth, discovery	1	G0012 Darkhotel	[32]
T1548.002	Bypass User Account Control	privilege-escalation	1	G0040 Patchwork	[32]
T1550.002	Pass the Hash	lateral-movement	1	G0050 APT32	[32]
T1550.003	Pass the Ticket	lateral-movement	1	G0050 APT32	[32]
T1552.002	Credentials in Registry	credential-access	1	G0050 APT32	[32]
T1555	Credentials from Password Stores	credential-access	1	G0038 Stealth Falcon	[32]
T1555.004	Windows Credential Manager	credential-access	1	G0038 Stealth Falcon	[32]
T1557	Adversary-in-the-Middle	credential-access, collection	1	G1041 Sea Turtle	[32]
T1560.001	Archive via Utility	collection	1	G1041 Sea Turtle	[32]
T1564.003	Hidden Window	stealth	1	G0050 APT32	[32]
T1564.004	NTFS File Attributes	stealth	1	G0050 APT32	[32]
T1564.011	Ignore Process Interrupts	stealth	1	G1041 Sea Turtle	[32]
T1566	Phishing	initial-access	1	G1041 Sea Turtle	[32]
T1566.003	Spearphishing via Service	initial-access	1	G0070 Dark Caracal	[32]
T1569.002	Service Execution	execution	1	G0050 APT32	[32]
T1570	Lateral Tool Transfer	lateral-movement	1	G0050 APT32	[32]
T1573	Encrypted Channel	command-and-control	1	G1002 BITTER	[32]
T1583	Acquire Infrastructure	resource-development	1	G1041 Sea Turtle	[32]

Technique	Name	Tactic	Groups (n)	Groups	Source
T1583.002	DNS Server	resource-development	1	G1041 Sea Turtle	[32]
T1583.003	Virtual Private Server	resource-development	1	G1041 Sea Turtle	[32]
T1584.002	DNS Server	resource-development	1	G1041 Sea Turtle	[32]
T1585.001	Social Media Accounts	resource-development	1	G0050 APT32	[32]
T1586.002	Email Accounts	resource-development	1	G0090 WIRTE	[32]
T1587.003	Digital Certificates	resource-development	1	G0056 PROMETHIUM	[32]
T1588.004	Digital Certificates	resource-development	1	G1041 Sea Turtle	[32]
T1589	Gather Victim Identity Information	reconnaissance	1	G0050 APT32	[32]
T1589.002	Email Addresses	reconnaissance	1	G0050 APT32	[32]
T1595.002	Vulnerability Scanning	reconnaissance	1	G0123 Volatile Cedar	[32]
T1595.003	Wordlist Scanning	reconnaissance	1	G0123 Volatile Cedar	[32]
T1608.003	Install Digital Certificate	resource-development	1	G1041 Sea Turtle	[32]
T1614	System Location Discovery	discovery	1	G1008 SideCopy	[32]
T1659	Content Injection	initial-access, command-and-control	1	G1019 MoustachedBouncer	[32]
T1684.001	Impersonation	stealth	1	G0090 WIRTE	[32]
T1685.005	Clear Windows Event Logs	defense-impairment	1	G0050 APT32	[32]
T1685.006	Clear Linux or Mac System Logs	defense-impairment	1	G1041 Sea Turtle	[32]
T1690	Prevent Command History Logging	defense-impairment	1	G1041 Sea Turtle	[32]

Mobile

The library's Mobile data records two techniques for Dark Caracal (T1437.001 Web Protocols, T1476 Deliver Malicious Application via Other Means), three for APT-C-23 (T1422 System Network Configuration Discovery, T1655.001 Match Legitimate Name or Location, T1660 Phishing) and two for the Mobile-only Bouncing Golf (T1476, T1655.001) (**Confirmed** [32]). The library generates no technique pages for the Mobile domain, so these are not linked. The Mobile software objects Pallas (S0399, Dark Caracal) and VajraSpy (S9006, Patchwork) have no library page. No group in this set has an ATT&CK for ICS record.

Groups, campaigns and software

Object	ID	Relationship in ATT&CK v20.0	Source
The 20 bound groups	G0050 G0121 G0040 G0142 G1002 G0134 G1008 G0012 G0038 G0070 G0123 G1005 G0021 G0090 G1028 G1041 G1019 G0056 G0063 G0089	Each linked in Part 1; every one has a generated page	[32]
Reviewed, not bound	G0078, G1035, G0126, G0095, G1013, G0055, G0020	Criminal mix, home in another study, or no state named	[10] [25] [27] [29] [30] [23] [31]
Bouncing Golf	G0097	Mobile only; no library page	[19]
Campaign C0011	C0011	Transparent Tribe against Indian students, December 2021 to July 2022	[8] [32]
Campaign C0033	C0033	PROMETHIUM StrongPity against Android users, May 2016 to January 2023	[22] [32]
UNC1151 / Ghostwriter	none	No object in the library's ATT&CK data	[32] [47]
Software	72 Enterprise IDs with library pages, plus the Mobile-only S0399 and S9006	See Part 8	[32]

CWE and CAPEC

KEV lists a CWE for 20 of the 22 CVEs in this set (Part 9), for example CWE-787 for CVE-2021-1732, CWE-119 for CVE-2017-11882, CWE-94 for CVE-2012-0158, CWE-78 for CVE-2014-6271, CWE-434 for CVE-2017-12617 and CWE-22 for CVE-2025-27920; only CVE-2017-0199 and CVE-2017-8570 have none (**Confirmed** [56]). CVE-2025-33053 carries CWE-73 (External Control of File Name or Path) in both KEV and NVD (**Confirmed** [51] [56]). CAPEC IDs come from the `capec` field of the library's `technique_profiles.jsonl` (**Confirmed** [32]). See the CWE Reference and CAPEC Reference.

Technique	CAPEC patterns linked in the library data
T1566.001 Spearphishing Attachment, T1566.002 Spearphishing Link, T1566.003 Spearphishing via Service (Dark Caracal)	CAPEC-163 Spear Phishing
T1078 Valid Accounts (Sea Turtle)	CAPEC-560 Use of Known Domain Credentials
T1690 Prevent Command History Logging (Sea Turtle)	CAPEC-13 Subverting Environment Variable Values; CAPEC-268 Audit Log Manipulation
T1584.002 DNS Server (Sea Turtle)	CAPEC-142 DNS Cache Poisoning
T1557 Adversary-in-the-Middle (Sea Turtle)	CAPEC-94 Adversary in the Middle (AiTM)
T1553.002 Code Signing (Darkhotel and others)	CAPEC-68 Subvert Code-signing Facilities; CAPEC-206 Signing Malicious Code; CAPEC-473 Signature Spoof
T1547.001 Registry Run Keys / Startup Folder	CAPEC-270 Modification of Registry Run Keys

Technique	CAPEC patterns linked in the library data
T1036.005 Match Legitimate Resource Name or Location	CAPEC-543 Counterfeit Websites; CAPEC-616 Establish Rogue Location
T1204.002, T1203, T1189, T1190	None in the library data

D3FEND countermeasures

The countermeasures below come from the library's `data/attack/technique_to_d3fend.jsonl` (D3FEND 1.6.0), and each links to its library page (**Confirmed** [32]). See the D3FEND Reference.

Behavior in this set	Techniques	D3FEND countermeasures in the library data
Spearphishing	T1566.001, T1566.002 (URL Analysis: T1566.002 only)	Email Filtering (D3-EF; D3FEND itself marks the relationship inferred and experimental [33]), URL Analysis (D3-UA), Dynamic Analysis (D3-DA)
Malicious files and run-key persistence	T1204.002, T1547.001	Executable Allowlisting (D3-EAL)
Client, privilege and server exploitation	T1203, T1068, T1190	Segment Address Offset Randomization (D3-SAOR), Process Segment Execution Prevention (D3-PSEP)
Run-key persistence	T1547.001	System Init Config Analysis (D3-SICA)
Web command and control	T1071.001	Outbound Traffic Filtering (D3-OTF), Certificate Analysis (D3-CA)
No countermeasure in the library data	T1584.002, T1588.004, T1553.002, T1659, T1685.006, T1690	See Part 13
Source guidance, not library-mapped	T1078, T1203, T1190, T1566.001, T1566.002	Multi-factor Authentication (D3-MFA), Software Update (D3-SU), Application Configuration Hardening (D3-ACH), as advised by [42] [43] [44] [45]

Techniques the major-state advisories state (comparison, not in the layer)

The government advisories summarized in Part 11 publish their own technique IDs. They are listed here for comparison and left out of this study's layer, because their home is the companion study for each state. Every ID was checked against the library's data (**Confirmed** [32]).

Advisory (date)	Actor	Techniques stated	Source
AA20-301A (2020-10-27)	Kimsuky	T1566.001, T1185, T1218.005, T1505.003, T1114.003 (among 27)	[94]
AA21-321A (2021-11-17)	Iranian government-sponsored APT	T1190, T1053.005, T1136.001, T1486	[100]
AA22-187A (2022-07-06)	North Korean state-sponsored actors (Maui)	T1059.008, T1486	[96]
AA23-040A (2023-02-09)	DPRK ransomware actors	T1190, T1133, T1195, T1486	[97]

Advisory (date)	Actor	Techniques stated	Source
AA23-129A (2023-05-09)	FSB Center 16 (Snake)	T1014, T1027.002, T1055.001, T1071.001, T1090.003, T1095, T1572, T1573	[86]
AA23-335A (2023-12-01)	IRGC-affiliated CyberAv3ngers	T1110, T1078.001, T1565.001, T1531, T1499, T1491.001	[103]
AA24-038A (2024-02-07)	Volt Typhoon	T1190, T1133, T1059, T1218, T1105, T1090.003, T1584.005 (among others)	[76]
AA24-190A (2024-07-08)	APT40	T1190, T1505.003, T1558.003, T1003, T1021.001, T1572 (among others)	[73]
AA24-207A (2024-07-25)	Andariel	T1190, T1003, T1021.002, T1567, T1572	[98]
AA24-290A (2024-10-16)	Iranian actors	T1110.003, T1621	[105]
AA25-141A (2025-05-21)	GRU Unit 26165	T1110.003, T1566.001, T1190, T1133, T1199, T1003.003	[87]
AA25-239A (2025-08-27)	Salt Typhoon-linked actors	T1190, T1199, T1098.004, T1686 (the advisory cites T1562.004, revoked in ATT&CK v20.0), T1599, T1602.001, T1602.002, T1572 (among 42)	[77]

The comparison shows the difference in kind. The big-five advisories lead with T1190 against edge devices, valid accounts and tunneling; this tier leads with T1204.002 and T1566.001. The shared ground is T1190 and T1566.001 (**Inferred** from both tables).

Navigator layer

The study's techniques are collected in an ATT&CK Navigator layer, navigator/library-layers/other-state-actors.json, listed with the library's other curated layers on the Navigator layers page. Open it in the ATT&CK Navigator. It holds the 154 Enterprise techniques above as 181 technique-tactic entries, domain enterprise-attack, ATT&CK version 20. The score is the number of the study's 20 groups that ATT&CK links to the technique (1 to 13); it measures breadth across the tier, not severity, and it is not an attribution claim. Each comment names the groups and the study's source numbers and, where a primary describes the technique, the dated report. Group lists in the comments follow ATT&CK's links only; the vendor-stated Stealth Falcon and UNC1151 entries appear under the comment's primary record, not as ATT&CK links for G0038.

Notices

The ATT&CK, CWE, CAPEC and D3FEND notices close the Sources part.

Cross-cutting themes

- **Attribution is a ladder, and this tier lives on the lower rungs.** One named agency, a handful of "suspected" countries, one explicitly unconfirmed link, and many groups with no state at all. A defender's report should carry the rung, not just the name. See the Threat Intelligence Reference.

- **Regional rivalry drives the targeting.** India-linked and Pakistan-linked groups target each other; Gaza groups target Israel; a Türkiye-linked actor targets Kurdish groups and governments opposing Turkish interests; a Belarus-aligned actor targets the embassies in its own capital (**Inferred** from Parts 2 to 5).
- **Civil society carries much of the cost.** Journalists, activists, dissidents, lawyers and diaspora communities appear in the Stealth Falcon, Dark Caracal, APT32 and BlackOasis records, often reached through their phones (**Confirmed** [11] [15] [35] [40]).
- **Bought tools blur the actor.** Commercial surveillance software and commodity remote access tools recur across groups, so tool hits are leads, not attributions (**Inferred** from Part 8).
- **Trust infrastructure is the outsized risk.** Registrars, registries, certificate issuance and the network path itself (Sea Turtle and MoustachedBouncer) sit beneath every other control (**Inferred** from [42] [48]).
- **The government spotlight falls elsewhere.** Joint advisories, sanctions and indictments concentrate on the big five; for this tier, vendors, researchers and platforms carry the public record (**Inferred** from Part 12).
- **The library's own data has edges worth knowing.** The library's Enterprise data holds 178 of the 182 live ATT&CK groups; the other four are ICS-only or Mobile-only and have no generated group page; Mobile techniques have no generated pages, and several of this tier's signature techniques have no D3FEND mapping (**Confirmed** [1] [32]).

Sources

Every source below was opened on 3 November 2026. Grades apply to the source type: Confirmed for a primary (the publisher of the fact), Reported for secondary coverage or for a leak publisher's or rival government's assertion.

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4	MITRE ATT&CK, Group G0121 Sidewinder	https://attack.mitre.org/groups/G0121/	2026-11-03	Confirmed
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#	Source	URL	Opened	Grade
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Related references

- Threat Actors
- Malware Families
- Notable Incidents
- CVE Reference
- Threat Intelligence Reference, Threat Hunting Reference
- Vulnerability Prioritization Reference, Vulnerability Management Reference
- Email Security Reference, Mobile Security Reference, Healthcare Security Reference
- CWE Reference, CAPEC Reference, D3FEND Reference
- Related case studies: All case studies; China-Attributed Threat Actors, Russia-Attributed Threat Actors, Operation Exchange Marauder, APT41: The Public Record, Stuxnet, Sandworm Operations, SolarWinds SUNBURST, Log4Shell, Change Healthcare 2024
- Generated MITRE pages: groups Sea Turtle `G1041`, APT32 `G0050`, Dark Caracal `G0070`, Transparent Tribe `G0134`, Patchwork `G0040`, Sidewinder `G0121`, Stealth Falcon `G0038`, MoustachedBouncer `G1019`, Darkhotel `G0012` and Equation `G0020`; campaigns `C0011` and `C0033`; software SnappyTCP `S1163`, Disco `S1088`, NightClub `S1090`, FinFisher `S0182`, BADNEWS `S0128` and Crimson `S0115`; techniques Malicious File `T1204.002`, Spearphishing Attachment `T1566.001`, DNS Server `T1584.002`, Content Injection `T1659` and Code Signing `T1553.002`